Probability Reliability and Quality

Probability, Statistics

Confidence Intervals

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Schedule and Learning Outcomes

Week	Recorded Lecture Topic
1	Introduction.
2	Inspection. Yield.
3	ANOVA. Measurement Systems Analysis.
4	**Statistics. Confidence Intervals**.
5	Hypothesis Testing. Sample Sizes.
6	Quality Management Processes.
7	Regression.
8	Design of Experiments.
9	System Reliability. Availability.
	BREAK – Non Teaching Week
10	Parametric Distribution Analysis
11	Statistical Process Control. Supplier Quality.
12	Review

After completion of the course the student should be able to:

- **Explain different statistical methods**
- **Parametric and non-parametric hypothesis tests**
- Design and plan for collecting experimental data
- Concepts of reliability engineering and the relationship between TTF, R, H
- How to improve its processes, products and services
- Apply TQM, 6S, and benchmarking

Motivation

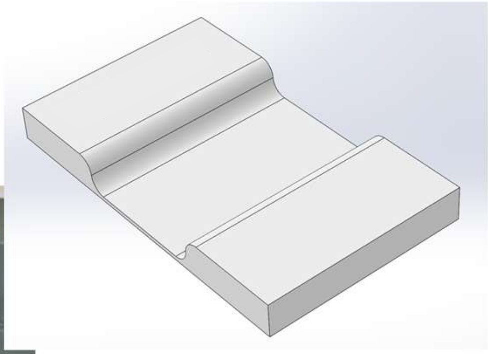
We fabricated parts according to dimensions & tolerances in CAD.

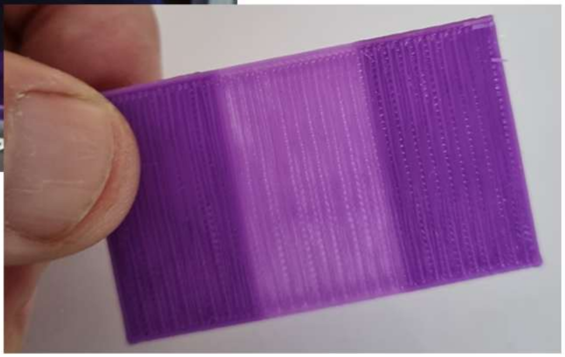
Did the printer make good parts?

Is the printing meeting nominal dimensions or are the parts way off?

How does the part-part variability compare to tolerance?

How confident are you that the printer make good parts tomorrow?





Motivation

In 5 years you will be responsible for managing the production of a new design.

You hire a remote supplier.

They show you a sample of units.

You inspect them.

How confident are you the supplier will make good units into the future?

Overview

Probability

Data Types

Distributions

Mean Statistic

Central Limit Theorem

Confidence Intervals

Probability

Probability is a way to quantify the belief of something being true or false.

- Tossing a coin and producing a “Head”? Pr = 0.5
- Roll a die and produce a “6”? Pr = 0.1667
- Will it rain tomorrow? Pr = 0.1
- Will it work? Pr = 0.8
- Is a unit tested within specification? Pr = 0.2

In manufacturing, you can speak about probability on a measured quantity.

Random Events

A **random experiment** is process that results in an outcome, but we are uncertain of what.

There is a **sample space** S of all possible outcomes

- An **outcome** is an element $s \in S$
- An **event** is a set A of outcomes, $A \subseteq S$

The set of all possible events is the set of all subsets of S, which includes the **certain** event S and the **impossible** event empty set $\emptyset$.

There may be elements outside of the space being sampled, but part of a larger set. That larger set is the universe or ***population*** of all elements.

A ***trial*** or ***run*** is an execution of the experimental process that results in a random outcome.

Random Events

Example: you fabricate a part.

It can meet specifications, or not meet specifications.

- The outcomes s are either "pass" or "fail".
- The sample space S = (pass, fail)
- Possible events are A = (pass), A = (fail), A = (pass ∪ fail), A = (pass ∩ fail) = ∅.

Example: you fabricate a part.

It can have no, one or more scratches.

- The outcomes s are either 0, 1, 2, … scratches (any non-negative integer).
- The sample space S = all non-negative integers.
- A possible event A might be A = (0) scratches, or A = (1-3) scratches, etc.

Example: you fabricate a part.

It has a certain length.

- The outcomes s are all possible lengths l, where $l \in \mathbb{R}$
- The sample space S = $\mathbb{R}$
- Possible events are intervals, for example A = (1.0, 1.1] or A = (1.0, 1.5] etc.

Random Events

Example: In the next 5 years, you will fabricate parts.

Parts can meet specifications or not.

- An outcomes s is a combination of “pass” or “fail”, for example s = {pass, pass, fail, pass, …}
- S = { (pass, fail), (pass, fail), (pass, fail), … }
- Possible events are combinations {pass, pass, fail, pass, …}, etc.

Example: In the next 5 years, you will fabricate parts.

Parts can have no, one or more scratches.

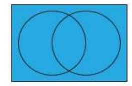
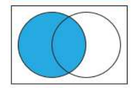
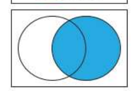
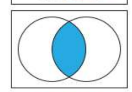
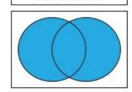
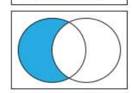
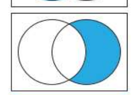
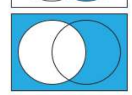
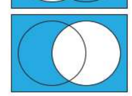
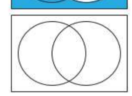
- An outcome s is a combination of # scratches
- The sample space S = $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \cdots$.
- Possible events A are {0, 0, 1, 0, 2, …} etc.

Example: In the next 5 years, you will fabricate parts.

Each has a certain length.

- An outcomes s is a set of lengths
- The sample space S = $\mathbb{R} \times \mathbb{R} \times \cdots$
- Possible events are { (0.9,0.95], (1.0, 1.1], …} etc.

Combining Events

Name	Notation	Definition	Venn diagram	Interpretation
Sample space	S	$\{x \in S : x \in S\}$		Certain event
Event	A	$\{x \in S : x \in A\}$		A occurs
Event	B	$\{x \in S : x \in B\}$		B occurs
Intersection	$A \cap B$	$\{x \in S : x \in A \text{ and } x \in B\}$		A and B occur
Union	$A \cup B$	$\{x \in S : x \in A \text{ or } x \in B\}$		A or B occurs (or both)
Difference	$A \setminus B$	$\{x \in S : x \in A \text{ and } x \notin B\}$		A occurs but B does not
Difference	$B \setminus A$	$\{x \in S : x \in B \text{ and } x \notin A\}$		B occurs but A does not
Complement	A^c	$\{x \in S : x \notin A\}$		A does not occur
Complement	B^c	$\{x \in S : x \notin B\}$		B does not occur
Empty set	$\emptyset$	$\{x \in S : x \notin S\}$		Impossible event

Probability Mathematics

A **probability** on a sample space S is a function from the events to a number $\in [0,1]$.

The probability of an event A is denoted Pr(A).

By definition, a probability function must follow:

1. The sample space S has probability Pr(S) = 1
2. For every event A we have $0 \leq \Pr(A) \leq 1$
3. For mutually exclusive events A, B, (where $A \cap B = \emptyset$), we have $\Pr(A \cup B) = \Pr(A) + \Pr(B)$

Independent Events

Two events A, B are **independent** if $P(A \cap B) = Pr(A) * Pr(B)$.

Otherwise they are dependent.

Sequential tosses of a coin or die are independent.

Sampling with replacement.

Repeated trials of a random experiment where a trial does not affect subsequent trials.

Repeated fabrication of a part (under what circumstances?).

Manufacturing Populations and Statistics

To understand and describe a ***population*** we collect ***samples*** and measure ***data***.

Population

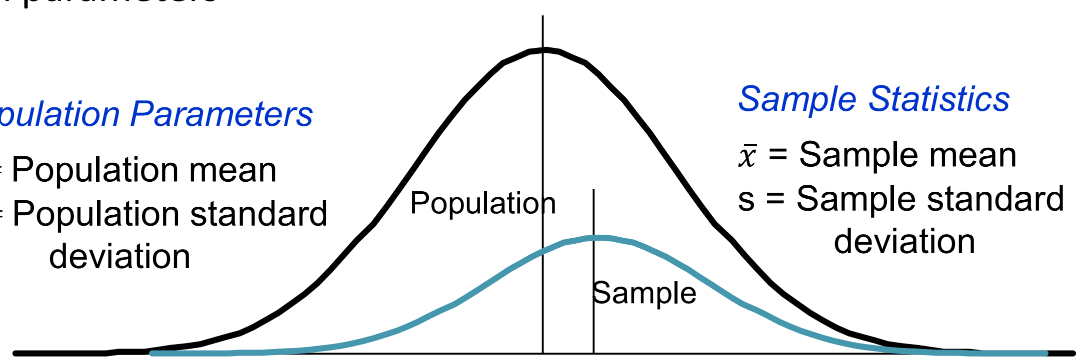
- A **population** is the entire group of objects that have been, could have been, or ever will be exist
- Each object is described with an outcome characteristic of interest
- A population **parameter** summarizes the population on the characteristic
- Population parameters are unknown and usually unknowable (projecting into the future)

Sample

- A **sample** is the group of objects actually measured in a study, a subset of the population of interest
- Measurements of the characteristic on each object forms the sample **data**.
- Sample **statistics** estimate population parameters

Population Parameters

μ = Population mean
σ = Population standard deviation



Sample Statistics

$\bar{x}$ = Sample mean
s = Sample standard deviation

Characteristics of Data

Type of Data

Shape

Measures of Central Tendency

Measures of Spread

Types of Numerical Data

<u>Attribute data</u>

- *Pass/fail* and *count of defects* are the most common attribute data
- Data we can count
  - Yes, No
  - Number of defects observed on a part
  - Pass / Fail: is it defective?

<u>Continuous data</u>

- Variable data, complete with units
- We can measure variables
  - Strip width
  - Press pressure
  - Process variables

Continuous data always provides more information than attributes and require lower sample sizes.


Pass/Fail

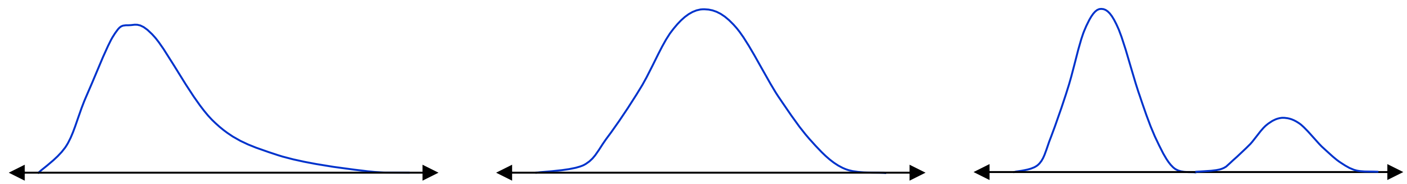

Count

Measure

Distribution of the Data

What does the occurrence data look like?

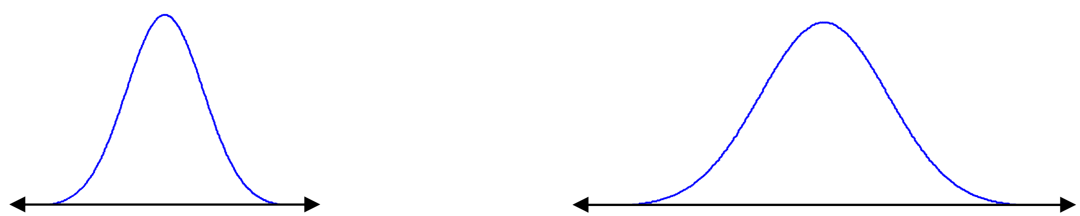
- Shape



- If the shape can be fit to a distribution (normal, Weibull, exponential, etc.) we call it **parametric**. If not, we call the data **non-parametric**.

- Central Tendency

- Variation



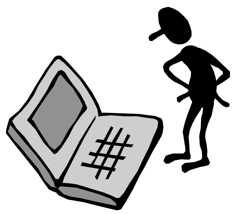
Analyzing Data

Graphical Analysis

- Run Charts
- Scatter Plots
- Box Plots
- Histograms

Descriptive Statistics

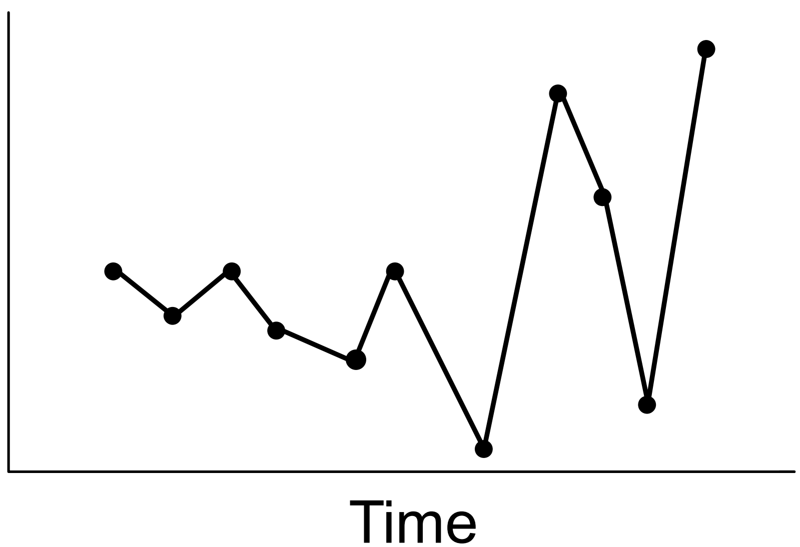
- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Range
- Percentiles



Run Chart

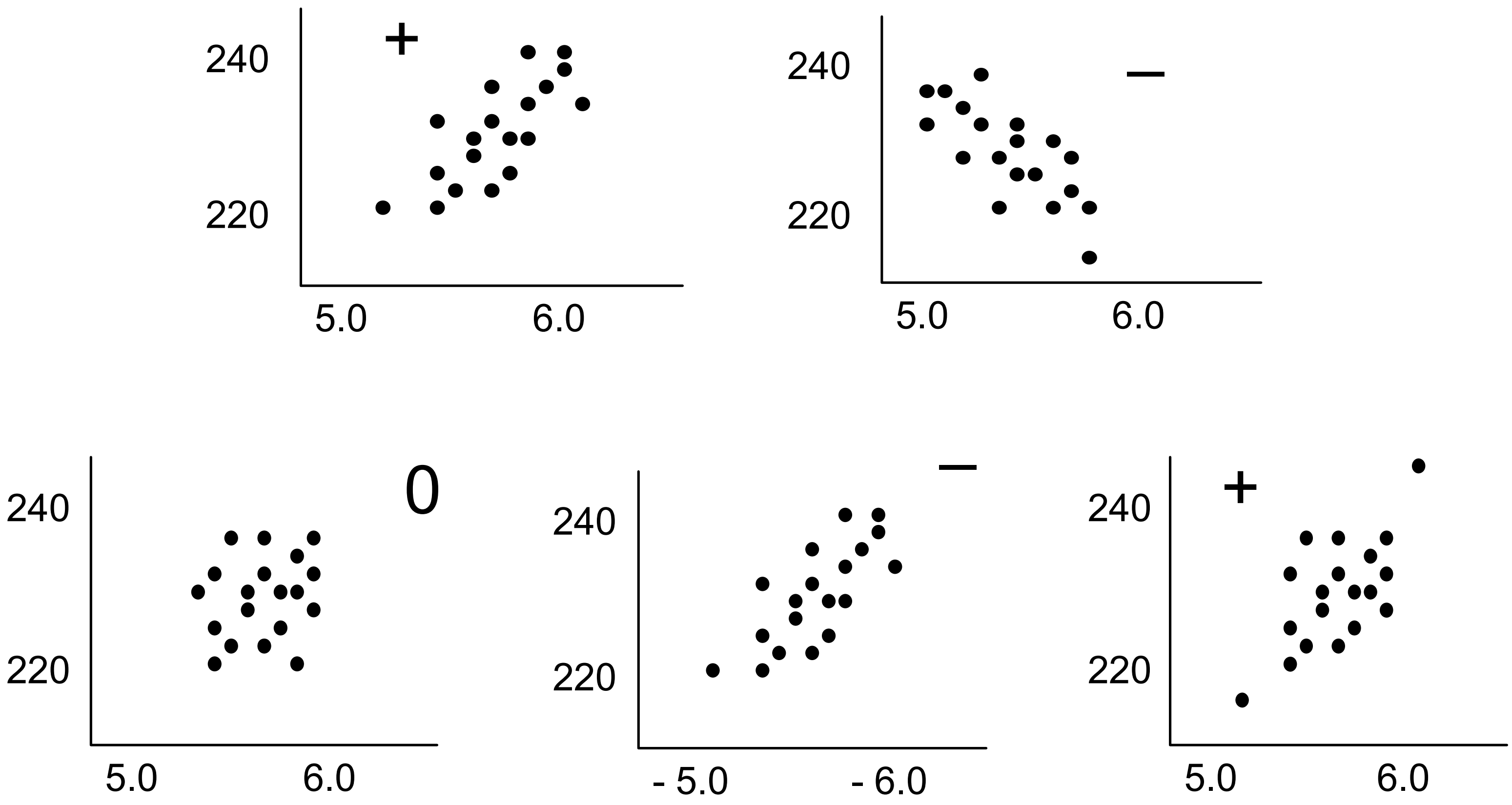
Helps to identify trends in data

Links changes in the process to points in time



Scatter Plots

Is there correlation? Positive, negative, none?



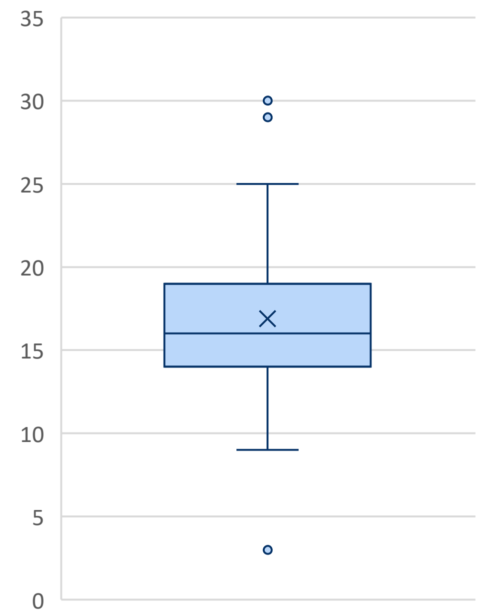
Boxplots

Shows the

- Range
- Interquartile range (1st quarter to 3rd quarter)
- Mean
- Median

Helps identify extreme values (outliers)

Useful for comparisons between multiple data sets

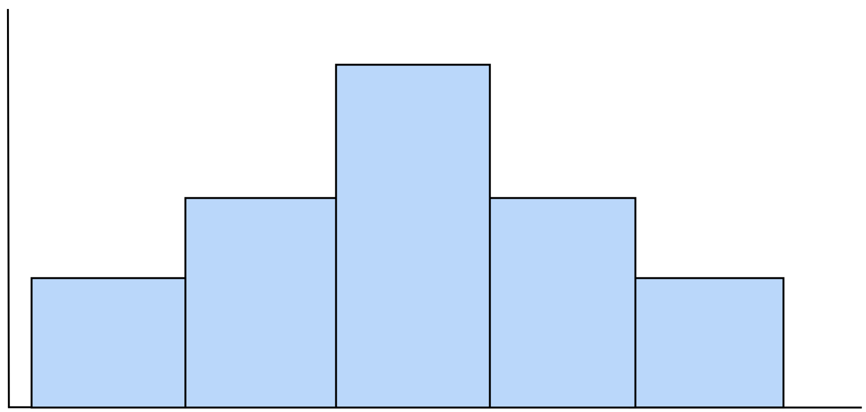


Histogram

When plotted against specifications, a histogram is a good way to view capability.

- Shows the relative frequency of occurrence of the various data values
- Reveals the centering, spread, and shape of the data

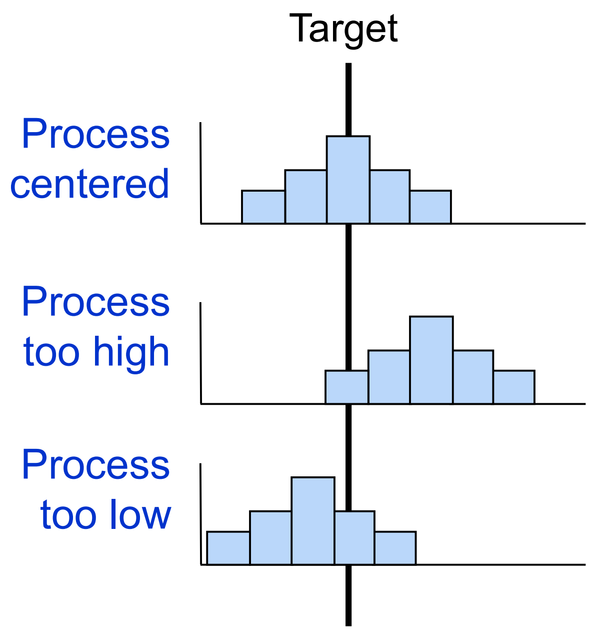
It can help answer the question, “Is the process capable of meeting requirements?”



Histogram: Assessing Process Capability

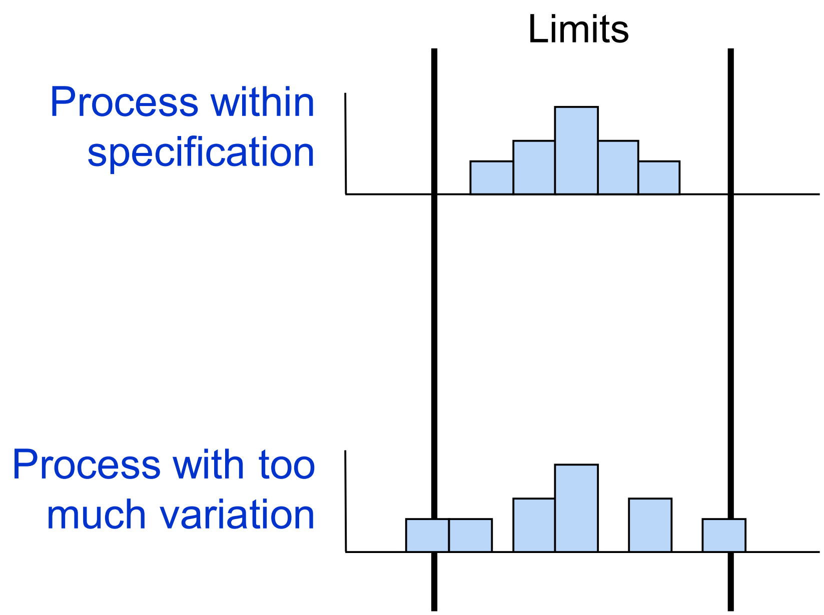
Central tendency –

Is the process distribution centered, too high, or too low?



Variation –

What is the spread of the data compared to the spec?



Analyzing Data

Graphical Analysis

- Run Charts
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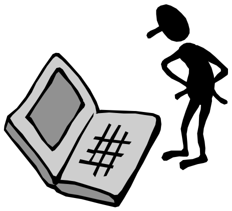
Descriptive Statistics

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Range
- Percentiles



Measures of Central Tendency

Mean: Arithmetic average of a dataset

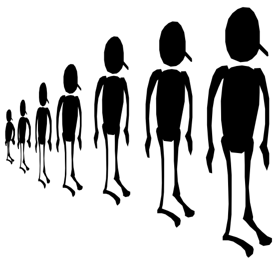
$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Influenced by ALL values of the dataset, including outliers

Median: half the sample data points are above the median, half below the median

- Not really influenced by extreme values or outliers
  - Sort all n values from the smallest to the largest
  - For odd n, the median is the (n+1)/2 sorted value;
    for even n, the median is the average of the (n/2)th and (n/2+1)th value

Mode: Represents the most frequent value of the data set



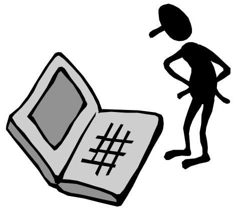
Analyzing Data

Graphical Analysis

- Scatter Plots
- Pareto Charts
- Run Charts
- Control Charts
- Box Plots
- Histograms

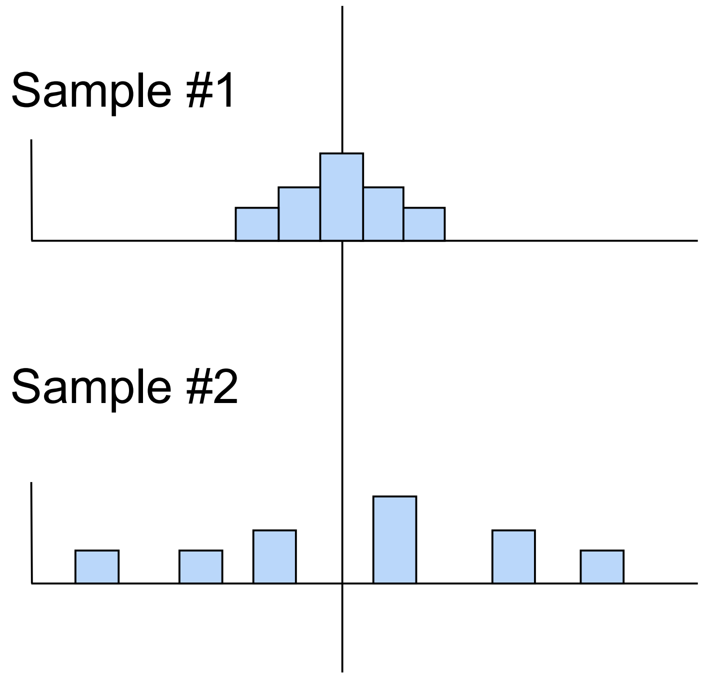
Descriptive Statistics

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Range
- Percentiles



Measures of Variability

Just knowing the location (or mean) of your data is not enough to properly describe your process.



Range

The ***Range*** R is the difference between the largest and smallest values of the data set

R = Largest Value – Smallest Value

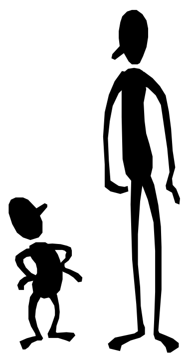
Advantages

- easy to understand and calculate

Disadvantages

- highly sensitive to extreme values or outliers.
- Only uses two values from the entire data set

The range can be defined for non-parametric data.

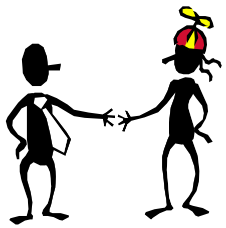


Sample Variance

The **variance** s^2 is the average of the squared deviations of each point from the sample average.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

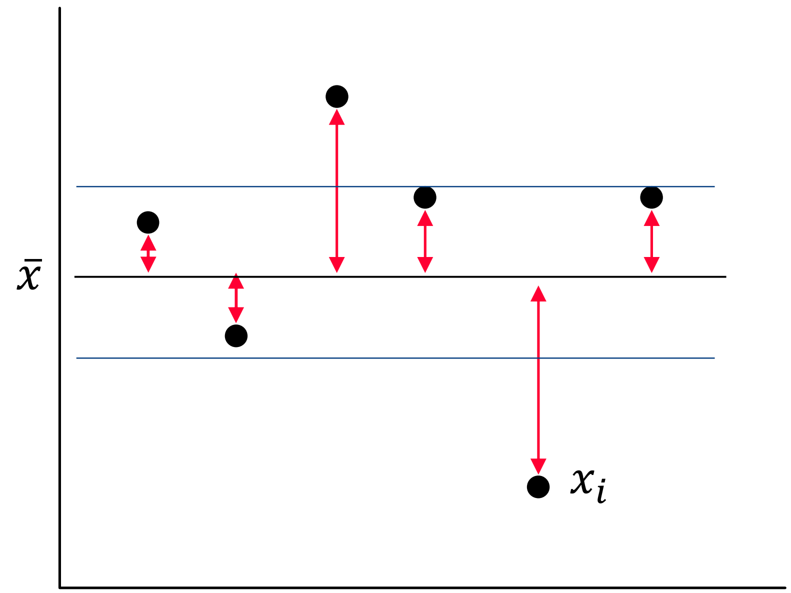
- The sample variance statistic s^2 is a sample estimate of the population standard deviation parameter σ^2.
- Advantage: additive estimate of the real variability
- Disadvantage: Variability gets measured in squared units, which can be confusing.



Sample Standard Deviation

The **standard deviation** s is the square root of the variance

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$



- Generally the best measure of variability
- Expressed in the same units as the original data, which helps its interpretation
- Uses all the data in its calculation, so can be sensitive to outliers.
- The sample variance statistic s is a sample estimate of the population standard deviation parameter σ.

Standard Deviation Math

Variances add, standard deviations do not.

Variances of the inputs add to calculate the total variance in the output

If σ^2_{total} = variance of the process output

$\sigma^2_{X_1}$ = variance due to Input Variable X_1

$\sigma^2_{X_2}$ = variance due to Input Variable X_2

Then $$\sigma^2_{\text{total}} = \sigma^2_{X_1} + \sigma^2_{X_2}$$

And $$\sigma_{\text{Total}} = \sqrt{\sigma^2_{X_1} + \sigma^2_{X_2}}$$

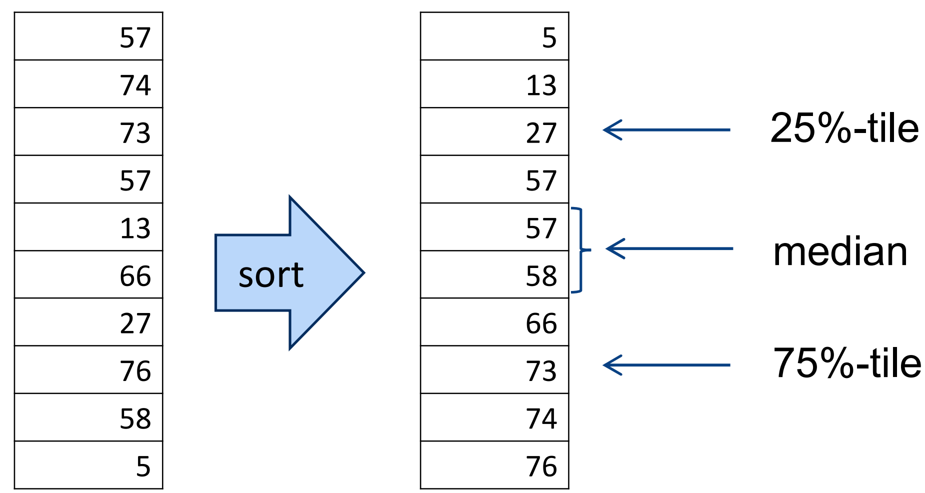
Quartile Range

The **25% quartile** is the value dividing the lower 25% of the data from the upper 75%.

The **75% quartile** is the value dividing the lower 75% of data from the upper 25%.

- Sort the n values of the data from smallest to largest.
- The n/4 value is the 25% quartile.
- The 3n/4 value is the 75% quartile.

The upper and lower quartile ranges are the differences from the median.



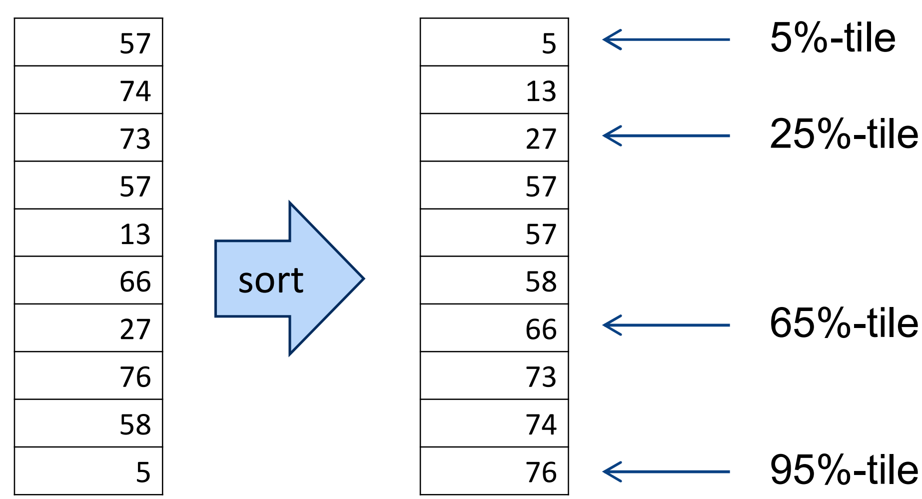
Percentiles

The **p-percentile** is the value dividing the lower p% of the data from the upper (1-p)%.

- Sort the n values of the sample from smallest to largest.
- The value of the pth point is the p-percentile.

For a sample, percentiles are defined only in increments of $\frac{1}{n}$, generally the values of $(\frac{1}{2n} + \frac{k}{n})$ for $k = [0, n-1]$.

The upper and lower "p-percentile ranges" are the differences from the median.



A Word on Outliers

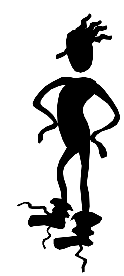
Points outside of expected results are ***outliers***.

- On a boxplot, those parts outside the whisker range.
- For normally distributed parts, those outside 3.5 to 4 sigma.

Outliers require special consideration

- They arose from non-typical but possible physics. How did they occur? Investigate.
- Those with known special causes can be removed from the data from statistical analysis
  - Treat outliers as a special, separate distribution, which is occurring X% of the time.
  - Treat the remainder of distribution as the usual distribution. Compute mean, standard deviation, probability plots, etc., using the “good” subset of the data and interpret as the base distribution

Do not include outliers in summary statistics calculations.
Rather, report as a “percent outlier” occurrence rate statistic.



Putting it All Together

With the histogram, form a summary slide with plots, statistics and confidence intervals.

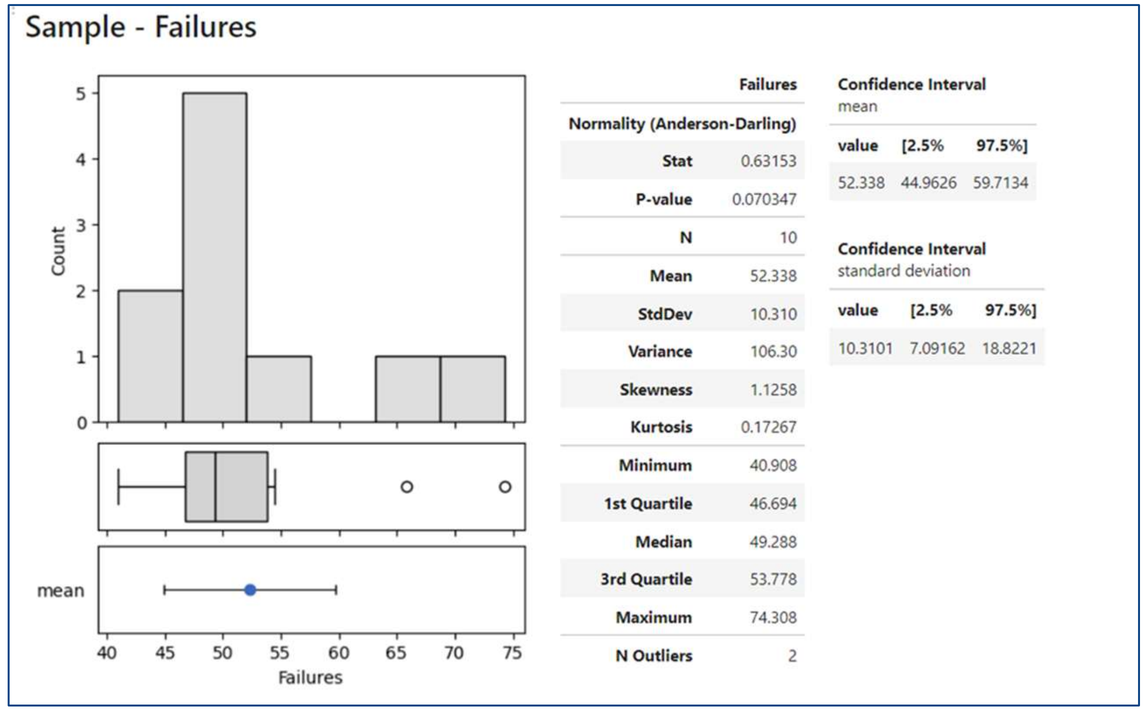
We can do this with Minitab or Python.

Minitab:

Stat > Basic Statistics > Graphical Summary

Python:

`Graphical Summary.ipynb` notebook



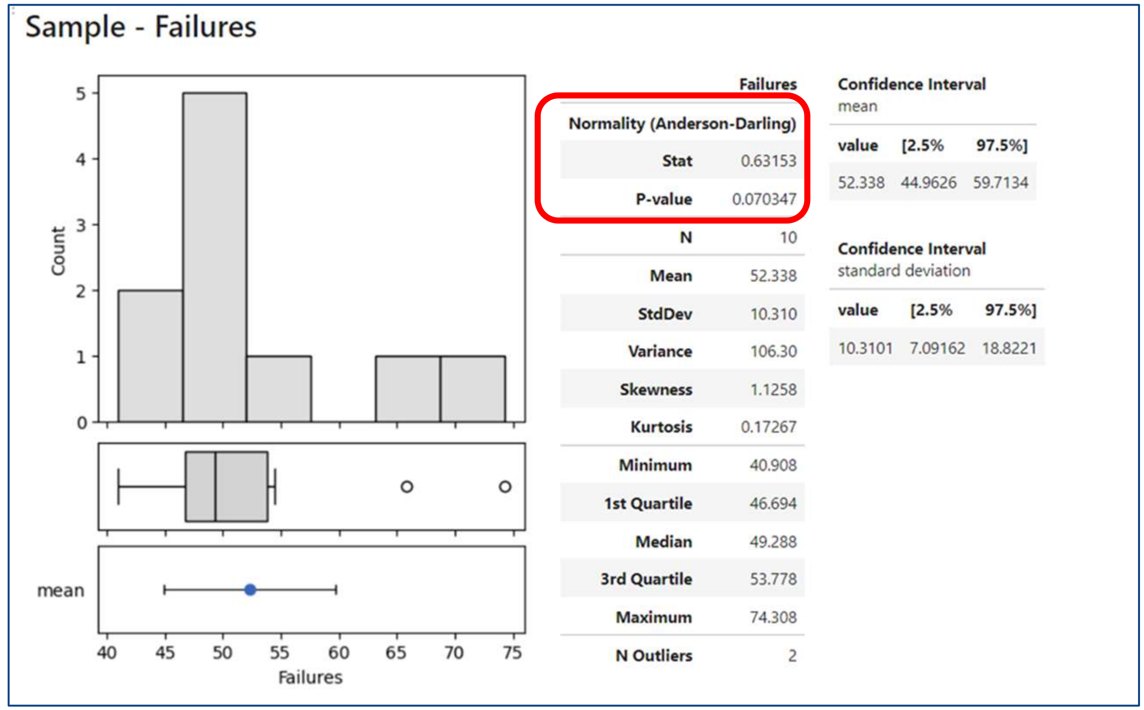
Normality Test

There is a test of a data set to indicate if the data comes from a normally distributed population, the ***Anderson-Darling test*** statistic.

The p-value of the test statistic indicates the probability the data's normal appearance was due to random events only, and is not really normal.

Failing the normality test with $p \le 0.05$ allows you to state with 95% confidence the data does not fit the normal distribution.

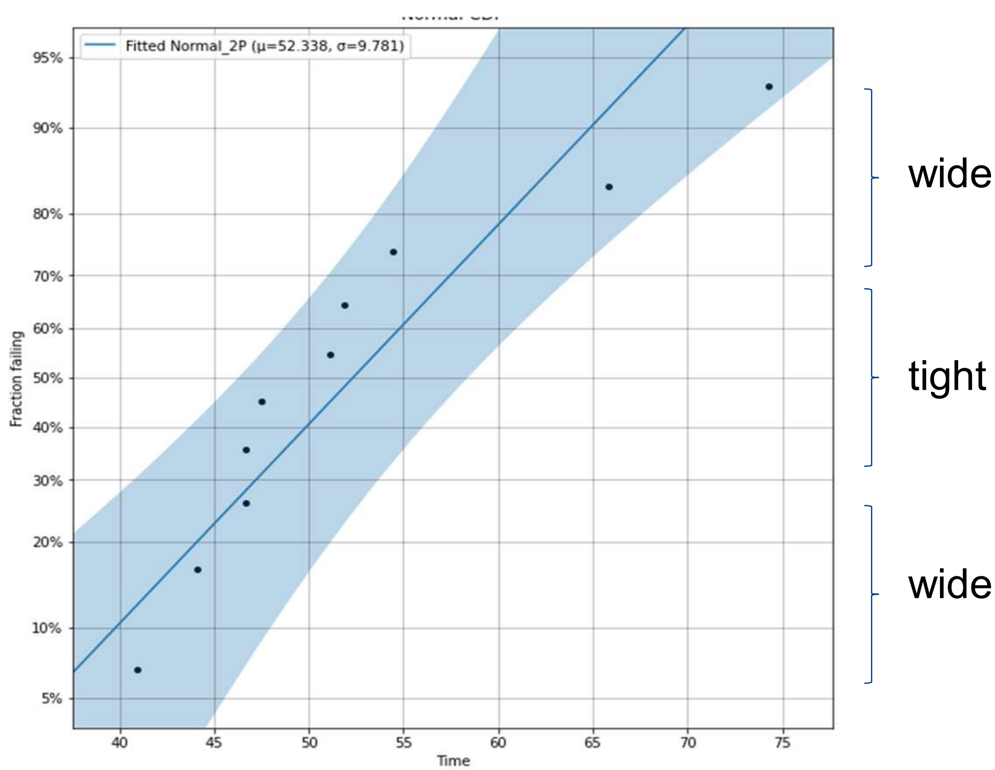
Here p is not less than 0.05, so we cannot say the data does not fit the normal distribution, but rather accept that it does.



Normal Probability Plot

A graphical version of the normality test is to plot the data on normal graph paper. Used to visually test whether a given data set can be described as "normal."

If a distribution is close to normal, the normal probability plot will be close to a straight line on normal probability plot graph paper.



Exercise: Population vs. Sample

Population	Sample
	10 hinges printed and provided in a sample 3 supplier parts randomly sampled every day this week 6 prototypes we test

Exercise: Population vs. Sample

Population	Sample
All 3D printed hinges that we may print this year, same material same settings	10 hinges printed and provided in a sample
All parts a supplier will provide, now and into the future	3 supplier parts randomly sampled every day this week
All future built units of a prototype design	6 prototypes we test

Measures of Central Tendency: Exercise

After inspecting 10 random production lots, the following number of defects were found on each of those lots

0, 4, 2, 1, 3, 2, 3, 1, 2, 2

Find the mean, median and mode of the data set

Mean =

Median =

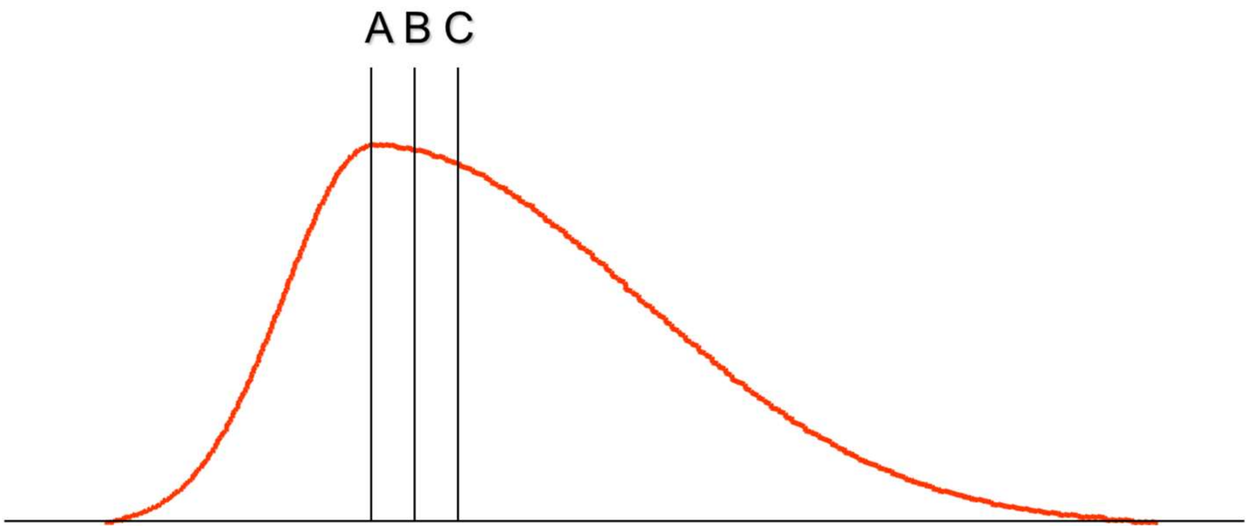
Mode =

What would happen if “4” is changed to “100”?

Measures of Central Tendency: Exercise

Three lines are represented in the following distribution.

Which is the mean, median and mode?



Putting it All Together

With the histogram, form a summary slide with plots, statistics and confidence intervals.

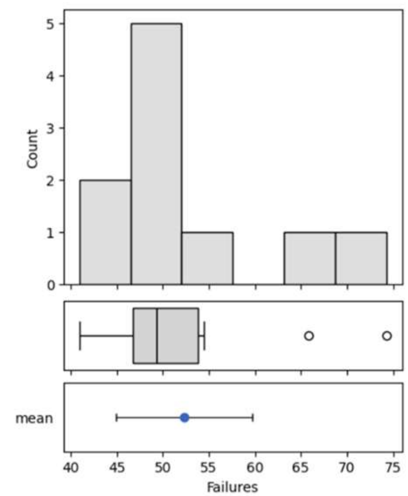
```
import mqr

data = np.array([54.41227487, 46.69129848, 74.30771187,
      47.4790787 , 51.09609842, 65.82481117, 40.90767595,
      44.08363342, 51.87603226, 46.70130042])

sample = mqr.summary.Sample(data, name='Failures’)

with Figure(4, 5, 3, 1, height_ratios=[4, 1, 1]) as (fig, ax):
    mqr.plot.summary(sample, ax=ax)
    summary = nb.grab_figure(fig)
nb.vstack(
    '## Sample - Failures',
    nb.hstack(
        summary,
        sample,
        nb.vstack(
            sample.conf_mean,
            sample.conf_std)))
```

Sample - Failures



	Failures
Normality (Anderson-Darling)	
Stat	0.63153
P-value	0.070347
N	10
Mean	52.338
StdDev	10.310
Variance	106.30
Skewness	1.1258
Kurtosis	0.17267
Minimum	40.908
1st Quartile	46.694
Median	49.288
3rd Quartile	53.778
Maximum	74.308
N Outliers	2

Confidence Interval
mean

value	[2.5%	97.5%]
52.338	44.9626	59.7134

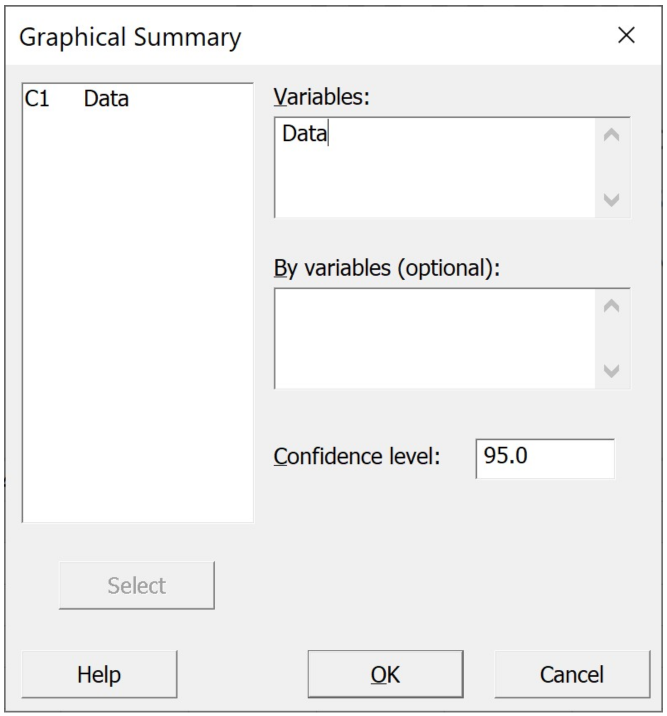
Confidence Interval
standard deviation

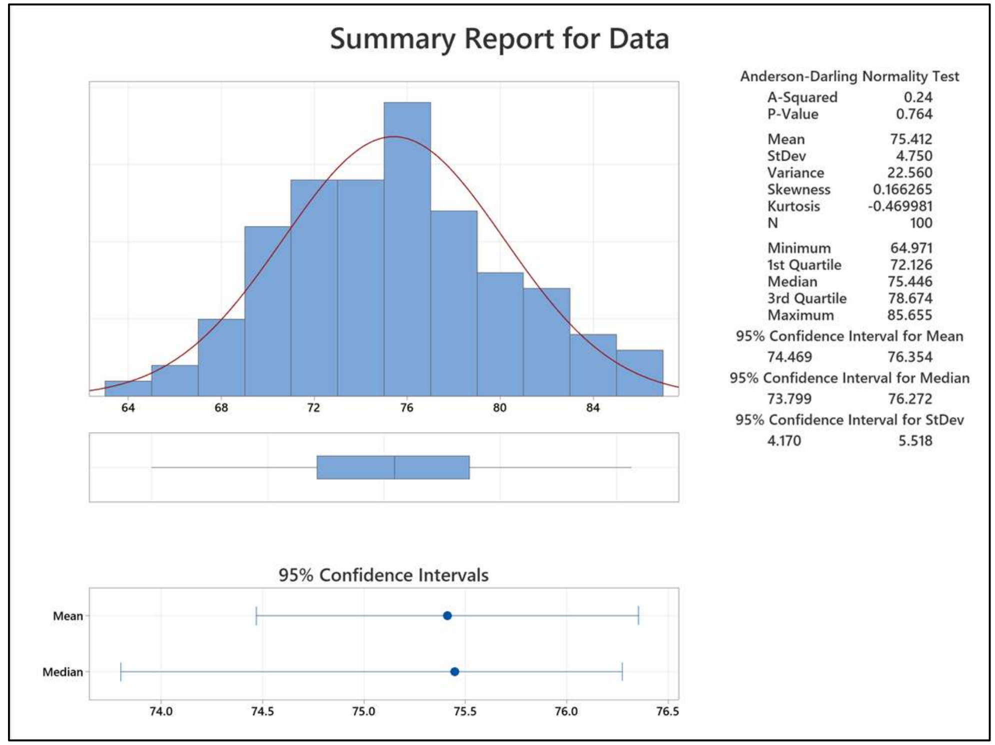
value	[2.5%	97.5%]
10.3101	7.09162	18.8221

Putting it all Together

In Minitab:

Stat > Basic Statistics > Graphical Summary





EXERCISE: Normal Probability Plot

Here is a sample of data.

Is it sampled from a normally distributed population?

```
54.41227487
46.69129848
74.30771187
47.47907870
51.09609842
65.82481117
40.90767595
44.08363342
51.87603226
46.70130042
```

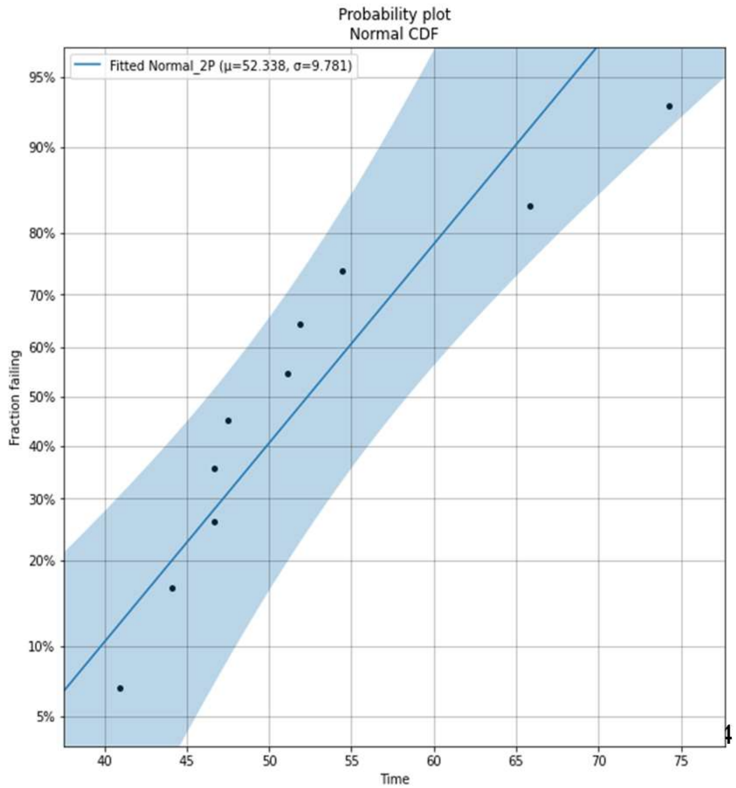
EXERCISE: Python: Normal Probability Plot

Suppose a process has generated parts with dimensions measured as follows:

54.41227487, 46.69129848, 74.30771187, 47.4790787 , 51.09609842, 65.82481117, 40.90767595, 44.08363342, 51.87603226, 46.70130042

Is the sample from a process that is normally distributed?

```
from reliability.Probability_plotting
        import Normal_probability_plot
import matplotlib.pyplot as plt
failures = [54.41227487, 46.69129848,
         74.30771187, 47.4790787 , 51.09609842,
         65.82481117, 40.90767595, 44.08363342,
         51.87603226, 46.70130042]
Normal_probability_plot(failures=failures)
plt.legend()
plt.show()
```



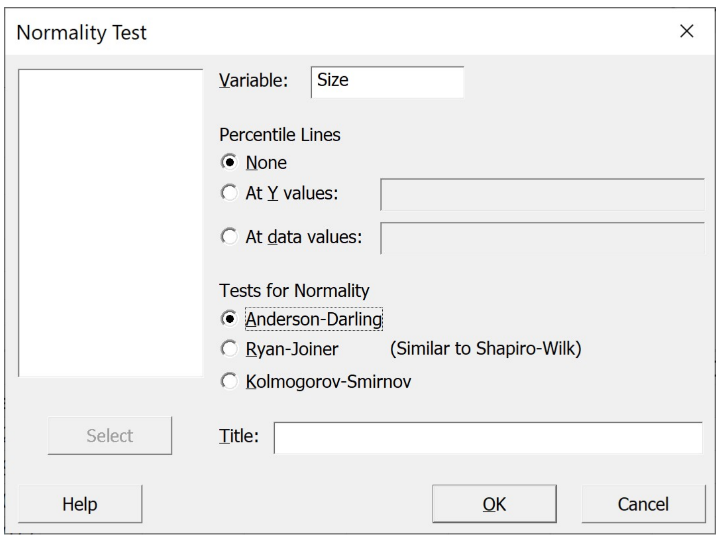
The data appears normal.

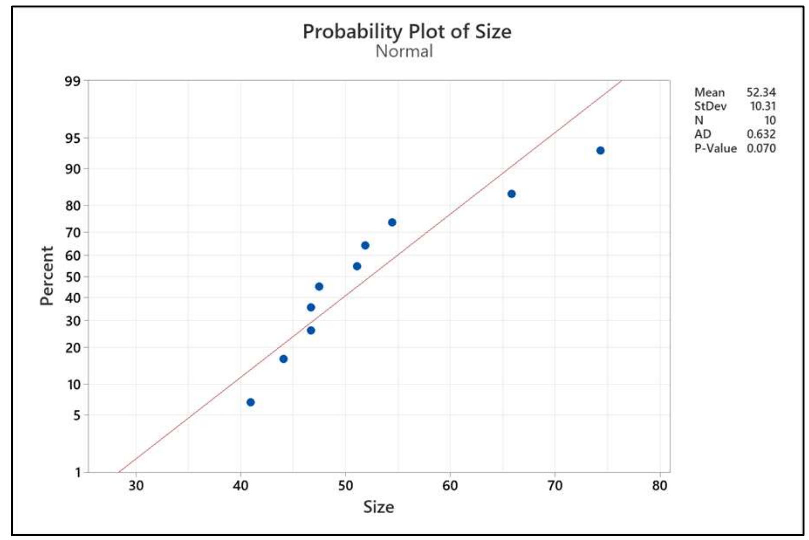
Minitab: Normal Probability Plot

Suppose a process has generated parts with dimensions measured as follows:

54.41227487, 46.69129848, 74.30771187, 47.4790787 , 51.09609842,
65.82481117, 40.90767595, 44.08363342, 51.87603226, 46.70130042

Is the sample from a process that is normally distributed?





The data appears normal.

Data Types

Suppose you make parts, and the visually inspect each part for shipment or reject scrap.

The (pass=1 fail=0) data is a binary set: {1,1,1,1,1,1,0,1,1,1,0,1,1,0,1,1,1,…}

Data Types

Suppose you make parts, and the visually inspect each part for shipment or reject scrap.

The (pass=1 fail=0) data is a binary set: {1,1,1,1,1,1,0,1,1,1,0,1,1,0,1,1,1,…}

Suppose you are painting the part, and you count the number of scratches on each part.

The data is the integer number of defects per part: {0,0,0,1,0,2,0,0,5,0,0,1,0,…}

Data Types

Suppose you make parts, and the visually inspect each part for shipment or reject scrap.

The (pass=1 fail=0) data is a binary set: {1,1,1,1,1,1,0,1,1,1,0,1,1,0,1,1,1,…}

Suppose you are painting the part, and you count the number of scratches on each part.

The data is the integer number of defects per part: {0,0,0,1,0,2,0,0,5,0,0,1,0,…}

Suppose you measure the length of the largest scratch on each part.

The data is the real valued number scratch length on each part: {0.0,0.0,0.0,0.4,0.0,1.2,…}

Data Types

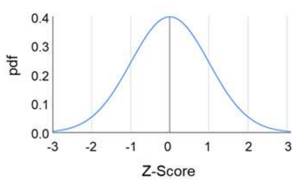
- Sample
  A dataset of numbers.
  - Continuous
    Real valued measurements of parts.
  - Discrete
    Judgements or counts on parts.
    - Pass/Fail
      Binary judgement.
      Defective unit or not.
    - Count
      Integer count.
      Number of defects per unit.

Data Types and Descriptive Distributions

Sample
A dataset of numbers.

Continuous
Real valued measurements of parts.

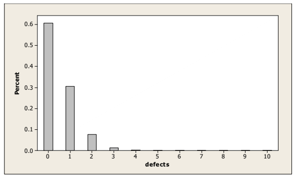
NORMAL



Discrete
Judgements or counts on parts.

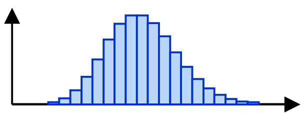
Pass/Fail
Binary judgement.
Defective unit or not.

BINOMIAL



Count
Integer count.
Number of defects per unit.

POISSON



Data Types and Distributions

Attribute

- Binomial: the distribution which governs counts of defectives
- Poisson: the distribution which governs counts of defects

Continuous

- Normal: the distribution which governs measured values from control & measurement errors

Additional

- Lognormal
- Exponential
- Weibull
- t
- Chi-square
- F

Binomial Distribution: Razor Example

A manufacturer produces disposable razors

- They make 100,000 per day. Historically, on average 500 are defective.
- To monitor quality of the process, a quality control technician chooses a daily sample of 100 randomly selected razors and records the number defective in the sample.

Data:

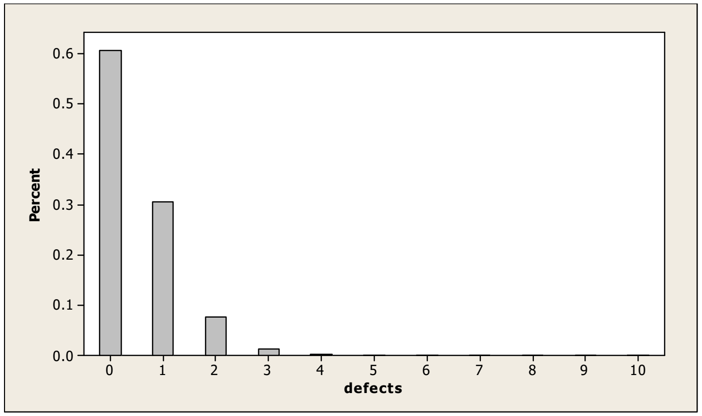
- The are 100 identical trials, $n = 100$
- Each trial has 2 possible outcomes, pass or fail
- $p = \frac{500}{100{,}000} = \ 0.005$ probability of finding a defective
- The number defective can be any number between 0 and 100.

Question:

- How likely will the daily sample show no defective, even when there is a 0.5% defect rate?
- How likely will the daily sample show 1 defective?
- How likely will the daily sample show less than 3 defective?

Binomial Distribution: Defectives

Consider using an inspection pass/fail test. For a set of inspections, the number of parts that result in a "fail" (or "pass") is binomially distributed when:



Conditions for a Binomial Experiment

- There are n identical trials
- Each trial has only 2 possible outcomes (pass/fail)
- The probability of each outcome remains constant for all trials
- The trials are independent

Characterized by

- n = sample size (number of trials)
- p = probability of a success or failure

The parameters are

- Central Tendency:
  Population Mean $\mu = N\pi$
- Spread:
  Population Variance $\sigma^2 = N\pi(1 - \pi)$

Binomial Probability Distribution

The binomial distribution function indicates the probability of a number of successes r of a number N of independent trials. It has two population parameters, N and π.

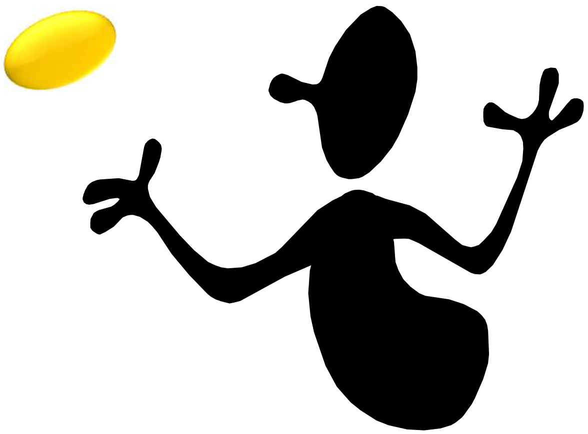
N = the population size

π = the probability of success

$$\Pr(r) = \frac{N!}{r!\,(N-r)!}\pi^r(1-\pi)^{N-r}$$

Binomial Probability Distribution

What is the probability of getting exactly 3 heads if a coin is flipped 6 times?



Binomial Probability Distribution

What is the probability of getting exactly 3 heads if a coin is flipped 6 times?

$N = 6$ and $\pi = 0.5$

$$\begin{aligned} \Pr(3) &= \frac{6!}{3!\,(6-3)!} 0.5^3 (1-0.5)^{6-3} \\ &= \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{(3 \cdot 2 \cdot 1)(3 \cdot 2 \cdot 1)} (0.125)(0.125) \\ &= 0.3125 \end{aligned}$$

There is a 31.25% chance of getting exactly 3 heads with 6 flips.

Binomial Probability Distribution

In Excel the binomial distribution function is given by

$Pr(X = r)$ =BINOM.DIST(N, π, r, FALSE)

And the cumulative distribution function is given by

$Pr(X \leq r)$ =BINOM.DIST(N, π, r, TRUE)

In Python the binomial distribution function is a `scipy` function

$Pr(X = r)$ = `binom.pdf(`r`,`N`,`π`)`

And the cumulative distribution function is given by

$Pr(X \leq r)$ = `binom.pmf(`r`,`N`,`π`)`

In Minitab, Calc > Probability Distributions > Binomial

Binomial Distribution: Razor Example

Back to the razor quality control.

$N = 100$ and $\pi = 0.005$

$$\Pr(0) = \frac{100!}{0!\,(100-0)!} 0.005^0 (1-0.005)^{100-0} = 0.606$$

$$\Pr(1) = \frac{100!}{1!\,(100-1)!} 0.005^1 (1-0.005)^{100-1} = 0.304$$

There is a 60% probability of a sample of 100 razors with of no defectives.

There is a 30% probability of 1 defective in a sample of 100 razors.

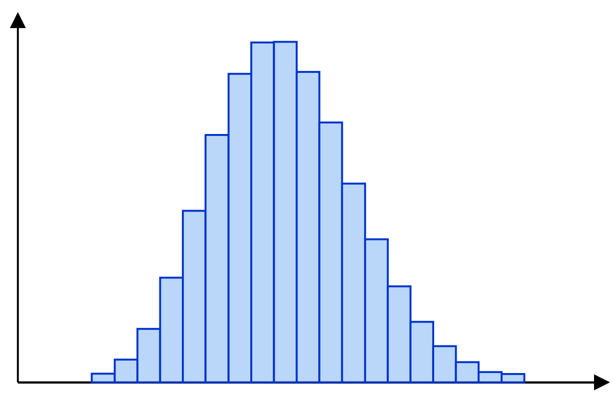
These probabilities decrease if the defective rate increases.

Poisson Distribution: Defects

Consider using an inspection that counts the number of defects on a part. For a set of inspections, the rate of defects per part are Poisson distributed. The distribution of the number of defect occurrences per unit interval (space, time, area, length, unit).

Examples:

- Number of visual defects on a part
- Number of flaws per square meter of a type of fabric
- Number of speeding tickets issued in a city in a week



Parameterized by

- λ = average number of occurrences per unit interval

Then distribution parameters are:

- Central tendency: mean $\mu = \lambda$
- Spread: variance $\sigma^2 = \lambda$

Poisson Calculation Example

An engineer has been studying the rate of paint spot defects arising a certain painting workstation.

- She collected data over a number of months, determined that the data fit the Poisson Distribution, and that the mean number of paint defects was 0.46 per unit.

She then wants to know what is the probability of having 2 or more spot defects on a unit.

Poisson Probability Distribution

The Poisson distribution function indicates the probability of a number x of independent events if they have a rate of occurrence λ. It has one population parameter the rate λ.

$$\Pr(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Poisson Probability Distribution

In Excel, the Poisson density function is

$Pr(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$ =POISSON.DIST(x, λ, FALSE)

And the cumulative probability function is

$Pr(X \leq x) = \sum_{t=0}^{x} \frac{\lambda^t e^{-\lambda}}{t!}$ =POISSON.DIST(x, λ, TRUE)

In Python, the Poisson density function is a `scipy` function

$Pr(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$ `= poisson.pdf(`x`,`λ`)`

And the cumulative probability function is

$Pr(X \leq x) = \sum_{t=0}^{x} \frac{\lambda^t e^{-\lambda}}{t!}$ `= poisson.pmf(`x`, `λ`)`

In Minitab, Calc > Probability Distributions > Poisson

Poisson Calculation Example

The probability of a unit have 2 or more defects is 1 minus the probability of zero or one defect.

$Pr(X \geq 2) = 1 - \Pr(X = 0) - \Pr(X = 1)$

$$Pr(X = 0) = \frac{0.046^0 e^{-0.046}}{0!} = 0.955$$

$$Pr(X = 1) = \frac{0.046^1 e^{-0.46}}{1!} = 0.0439$$

Therefore $Pr(X \geq 2) = 1 - 0.955 - 0.0439 = 0.001$

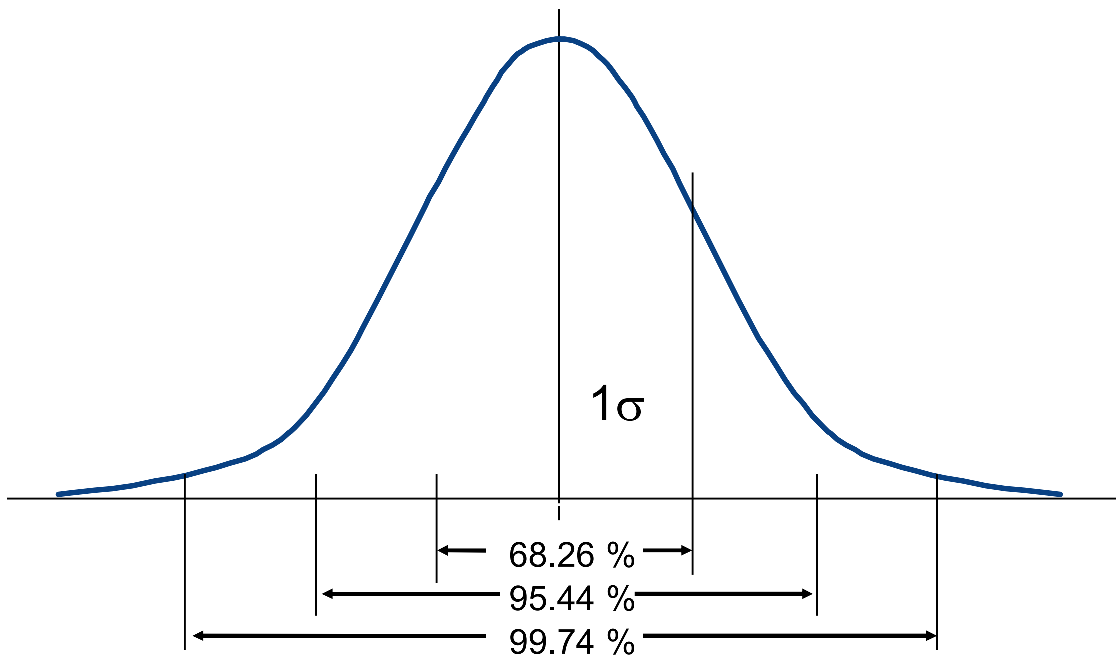
The probability of a unit with two or more spot paint defects is about 0.1%.

Normal Distribution

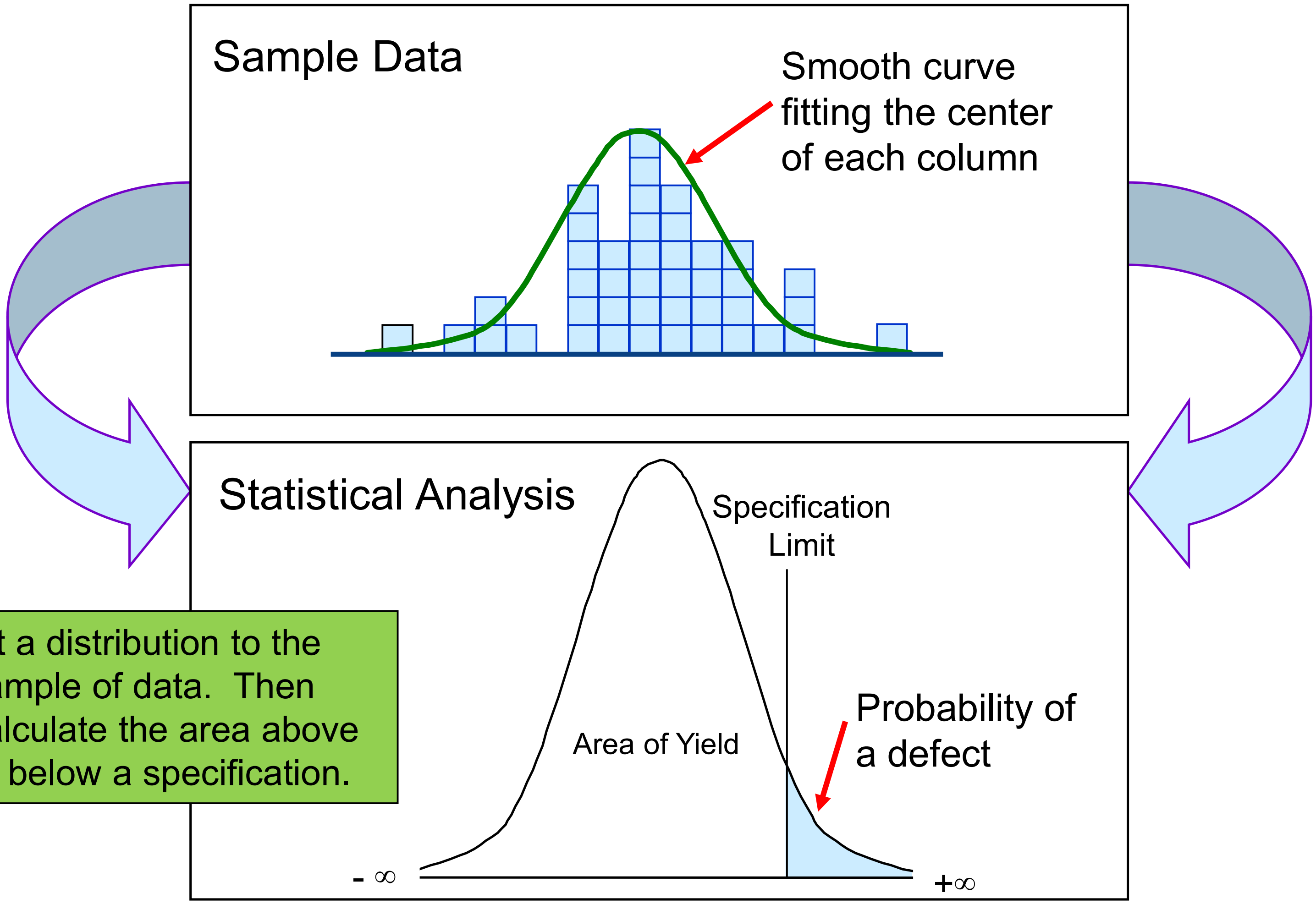
Consider using an inspection that measures a length dimension. The distribution of the length occurrences will often be normal.

Parameterized by

- μ = the average
- σ = the standard deviation



Normal Distribution



Fit a distribution to the sample of data. Then calculate the area above or below a specification.

Normal Probability Distribution

Consider a population of observations whose values are real-valued and normally distributed with population mean μ and standard deviation σ.

Probability is defined by the cumulative distribution function Φ, where

$$\Phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

From this, the probability of an event as an observation within an interval is

$$\Pr\big((a,b)\big) = \Phi(b) - \Phi(a) = \frac{1}{\sigma\sqrt{2\pi}} \int_{a}^{b} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Also, the probability density function is defined by

$$\varphi(x) = \frac{d\Phi}{dx} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Normal Probability Distribution – Excel

In Excel, the normal distribution density function is

$\varphi(x) =$ =NORM.DIST(x, μ, σ, FALSE)

And the cumulative distribution function is

$\Phi(X \leq x) =$ =NORM.DIST(x, μ, σ, TRUE)

And the probability of an event interval is

$\Pr((a, b)) =$ =NORM.DIST(b, μ, σ, TRUE)-NORM.DIST(a, μ, σ, TRUE)

Similarly, given a probability value, the cumulative inverse function Z is

$x = \Phi^{-1}(Pr) =$ =NORM.INV(Pr, μ, σ)

Normal Probability Distribution – Python

In Python, the normal distribution is a `scipy` function

$\varphi(x) =$ = `norm.pdf(`x, μ, σ`)`

And the cumulative distribution function is

$\Phi(X \leq x) =$ = `norm.cdf(`x, μ, σ`)`

And the probability of an event interval is

$\Pr((a, b)) =$ = `norm.cdf(`b, μ, σ`) - norm.cdf(`a, μ, σ`)`

Similarly, given a probability value, the cumulative inverse function Z is

$x = \Phi^{-1}(Pr) =$ =`norm.ppf(Pr,`μ, σ`)`

Computational Equations

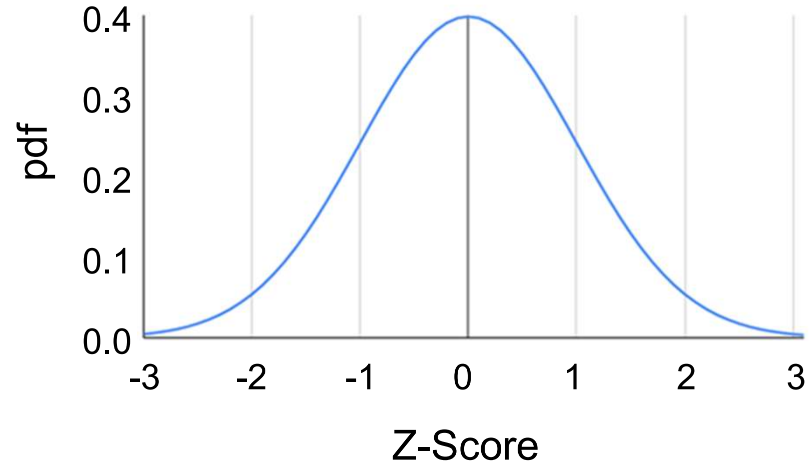
When the normal distribution has $\mu = 0$, $\sigma = 1$, it is known as a ***standard normal*** distribution, and is denoted by ***Z***.

Any normal distribution X can be transformed to a standard normal distribution using the ***Z-score***

$$Z = \frac{x - \mu}{\sigma}$$

An x value's Z-score is
"the number of sigmas away from the mean"

The Z-score is the fundamental yardstick used to evaluate the measured performance of a process and comparing different processes.
It is directly related to the probability of defectives.



Computational Equations

Historically, the standard normal distribution was tabulated and used to compute the any normal distribution values. Compute the Z value, then look up cumulative probabilities.

Now we have statistical software.

	Excel	Python
$\varphi(Z)$	=NORM.S.DIST(Z, FALSE)	`=norm.pdf(Z,0.0,1.0)`
$\Phi(z \le Z)$	=NORM.S.DIST(Z, TRUE)	`=norm.cdf(Z,0.0,1.0)`
$Z = \Phi^{-1}(Pr)$	=NORM.S.INV(Pr)	`=norm.ppf(Pr,0.0,1.0)`

Z-Score Table

Cumulative probabilities for NEGATIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

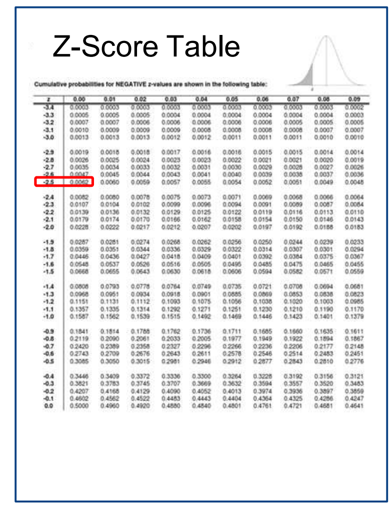
Example

A metal bar manufacturing process is found to be normally distributed with a mean of 20 inches and a standard deviation of 0.1 inches. The spec limit is 20.25 inches maximum.

Estimate the amount of product manufactured outside spec.

- mean = 20 inches
- stdev = 0.1 inches

$$Z = \frac{x - \mu}{\sigma} = \frac{20.25 - 20}{0.1} = 2.5$$



From the cumulative distribution Z-score tabled data, Pr(Z < 2.5) = 0.9938.

So Pr(X > 20.25) = Pr(Z > 2.5)
= 1 – Φ(Z < 2.5)
= 1 - 0.9938 = 0.0062 or 0.62%

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A metal bar manufacturing process is found to be normally distributed with a mean of 20 inches and a standard deviation of 0.1 inches. The spec limit is 20.25 inches maximum.

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$$Z = \frac{x - \mu}{\sigma} = \frac{20.25 - 20}{0.1} = 2.5$$

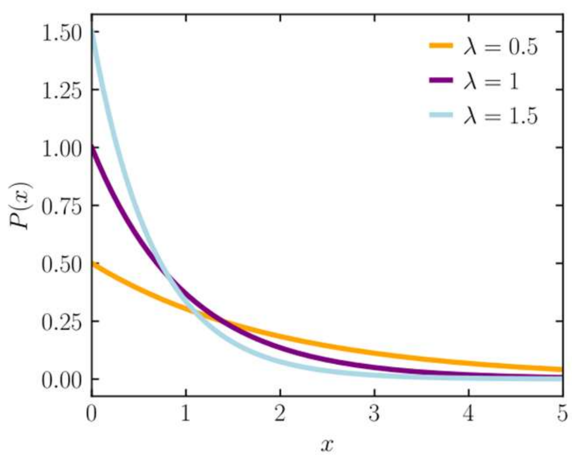
From the cumulative distribution Z-score tabled data, $\Pr(Z < 2.5) = 0.9938$.

So $\Pr(X > 20.25) = \Pr(Z > 2.5)$
$= 1 - \Phi(Z < 2.5)$
$= 1 - 0.9938 = 0.0062$ or 0.62%

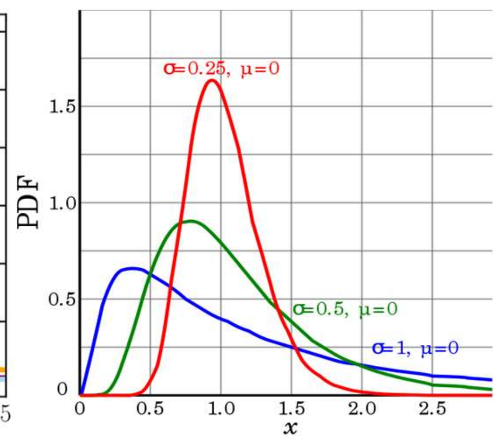
We expect to manufacture 0.62% of parts outside specification.

Common Reliability Distributions

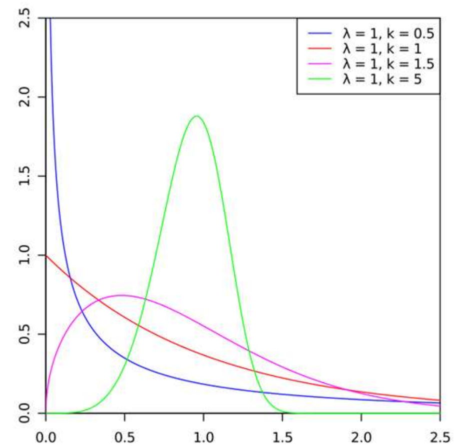
Three distributions frequently occur in reliability situations:



Exponential



Lognormal



Weibull

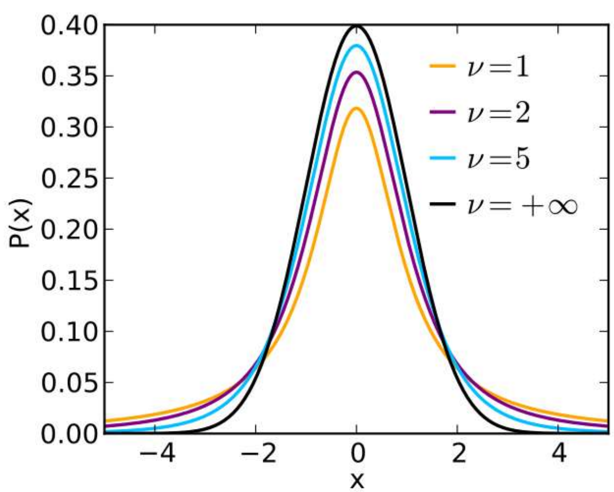
These distributions may also find use in waiting time applications, such as cycle times or queue times.

Common Sampling Distributions

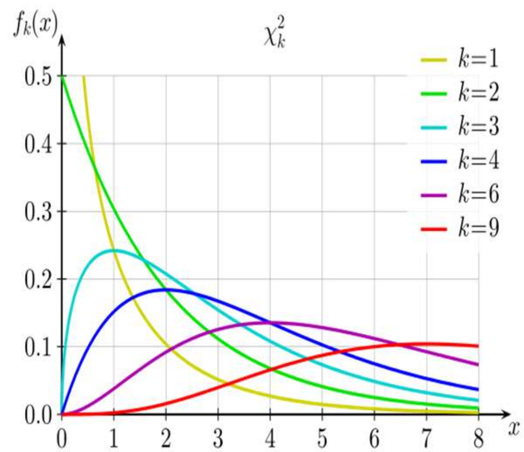
Different statistics of interest – such as means, standard deviations, rates and proportions – are distributed in particular ways.

Knowledge of these distributions allows the estimation of the probability of occurrence

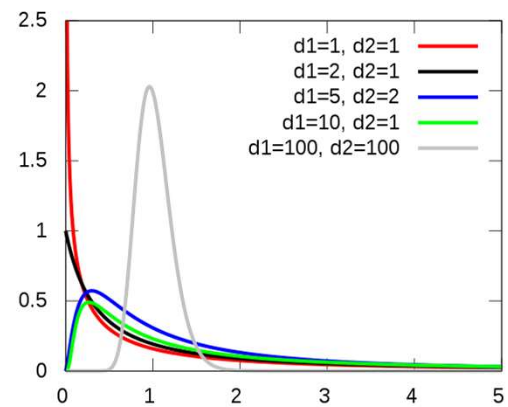
Important distributions related to statistics of interest include:



t: distribution of sample means



Chi-squared: distribution of sample variances



F: distribution of ratio of two sample variances

Useful Distribution Functions Summary

Distribution	Typical Use	Math	Excel	Python
Binomial	Pass/fail data	$Pr(X = r)$	BINOM.DIST(N, π, r, FALSE)	`binom.pdf` (r, N, π)
		$Pr(X \leq r)$	BINOM.DIST(N, π, r, TRUE)	`binom.pmf` (r, N, π)
		$Pr^{-1}(p)$	BINOM.INV(N, π, p)	`binom.inv` (p, N, π)
Poisson	Defect count data	$Pr(X = r)$	POISSON.DIST(x, λ, FALSE)	`poisson.pdf` (r, N, π)
		$Pr(X \leq r)$	POISSON.DIST(x, λ, TRUE)	`poisson.pmf` (r, N, π)
		$Pr^{-1}(p)$	POISSON.DIST(p, λ)	`poisson.inv` (p, N, π)
Normal	Measurement error	$\varphi(x)$	NORM.DIST(x, μ, σ, FALSE)	`norm.pdf` (x, μ, σ)
		$\Phi(X \leq x)$	NORM.DIST(x, μ, σ, TRUE)	`norm.cdf` (x, μ, σ)
		$\Phi^{-1}(Pr)$	NORM.INV(Pr, μ, σ)	`norm.ppf` (x, μ, σ)

In Minitab, these are all under Calc > Probability Distributions

Summary

The binomial distribution governs sampling and estimation of defectives

The Poisson distribution governs sampling and estimation of defects

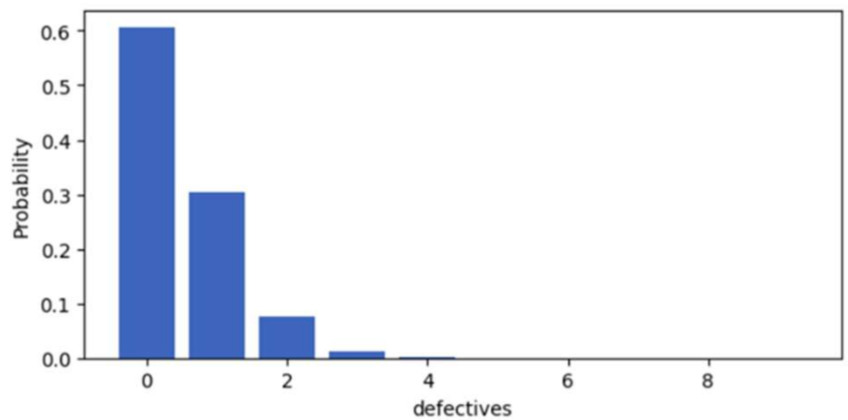
The normal distribution governs sampling and estimation of measured values arising from measurement and control limited processes

Other distributions are appropriate for processes governed by other physics.

We will focus on these three in this course for manufacturing quality analysis

EXERCISE: Binomial Distribution: Razor Example

```python
from scipy.stats import binom
import mqr

n = 100
p = 0.005
r_values = list(range(10))

dist = binom(n, p)
pmf_values = dist.pmf(r_values)

df = pd.DataFrame(
    {'Pr(r)': pmf_values},
    index=pd.Index(r_values, name='r'))

# display table and statistics
stats = f'''
### Stats
mean: {dist.mean():.3g}
variance: {dist.var():.3g}
'''

with Figure() as (fig, ax):
    ax.bar(r_values, pmf_values)
    ax.set_xlabel('defectives')
    ax.set_ylabel('Probability')
    plot = nb.grab_figure(fig)

nb.hstack(plot, df, stats)
```



r	Pr(r)
0	6.057704e-01
1	3.044073e-01
2	7.571939e-02
3	1.242965e-02
4	1.514668e-03
5	1.461388e-04
6	1.162746e-05
7	7.846242e-07
8	4.583546e-08
9	2.354474e-09

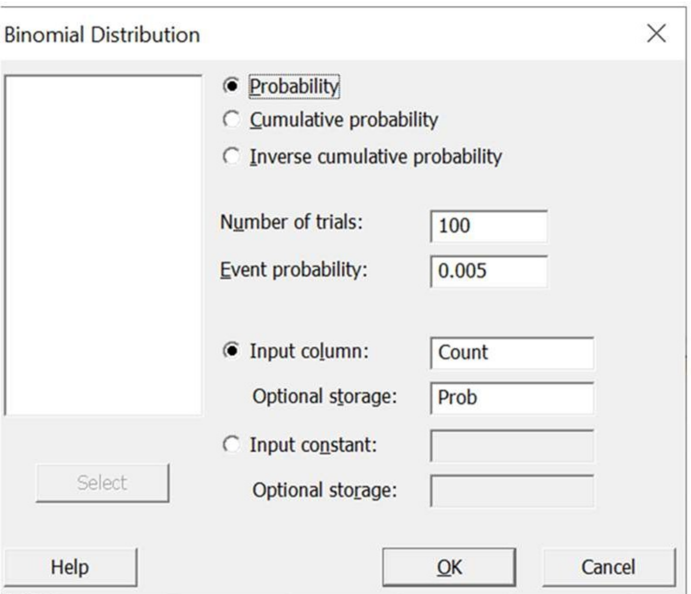
Stats

mean: 0.5

variance: 0.497

Binomial Distribution: Razor Example

Calc > Probability Distributions > Binomial

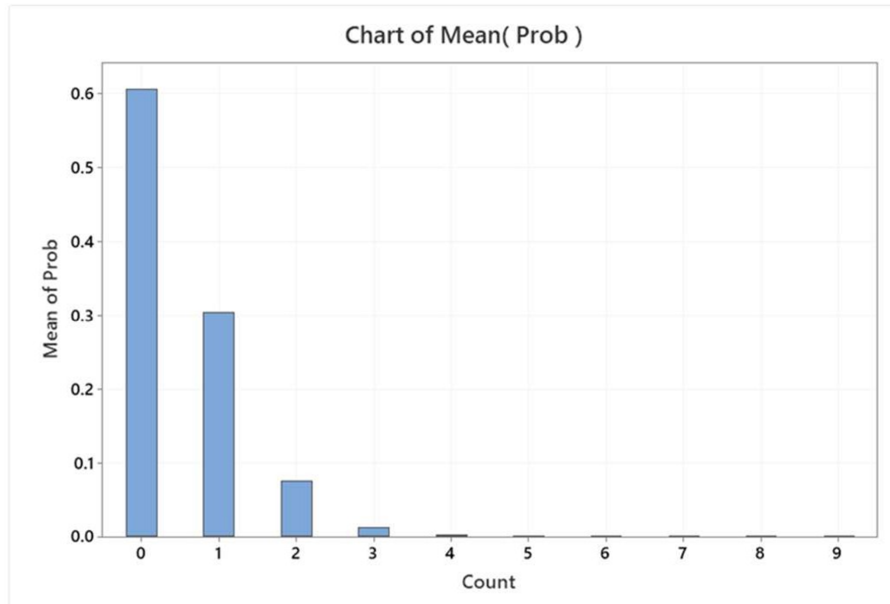


Binomial with n = 100 and p = 0.005

x	P(X = x)
0	0.605770
1	0.304407
2	0.075719
3	0.012430
4	0.001515
5	0.000146
6	0.000012
7	0.000001
8	0.000000
9	0.000000

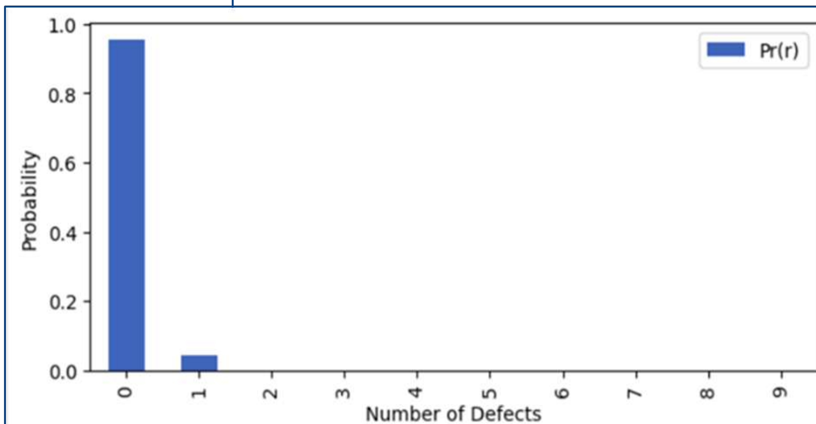
C3	C4
Count	Prob
0	0.605770
1	0.304407
2	0.075719
3	0.012430
4	0.001515
5	0.000146
6	0.000012
7	0.000001
8	0.000000
9	0.000000

Worksheet 1



EXERCISE: Poisson Calculation Example

```
from scipy.stats import poisson
import matplotlib.pyplot as plt

lam = 0.046
r_values = list(range(10))

dist = poisson(lam)
Pmf_values = poisson.pmf(r_values)
df = pd.DataFrame({'Pr(r)': pmf_values},
         index=pd.Index(r_values, name='r'))
stats = f'''
### Stats
mean: {dist.mean():.3g}
variance: {dist.var():.3g}
'''
with Figure() as (fig, ax):
    df.plot.bar(ax=ax)
    ax.set_xlabel('Number of Defects')
    ax.set_ylabel('Probability')
    plot = nb.grab_figure(fig)

nb.hstack(plot, df, stats)
```



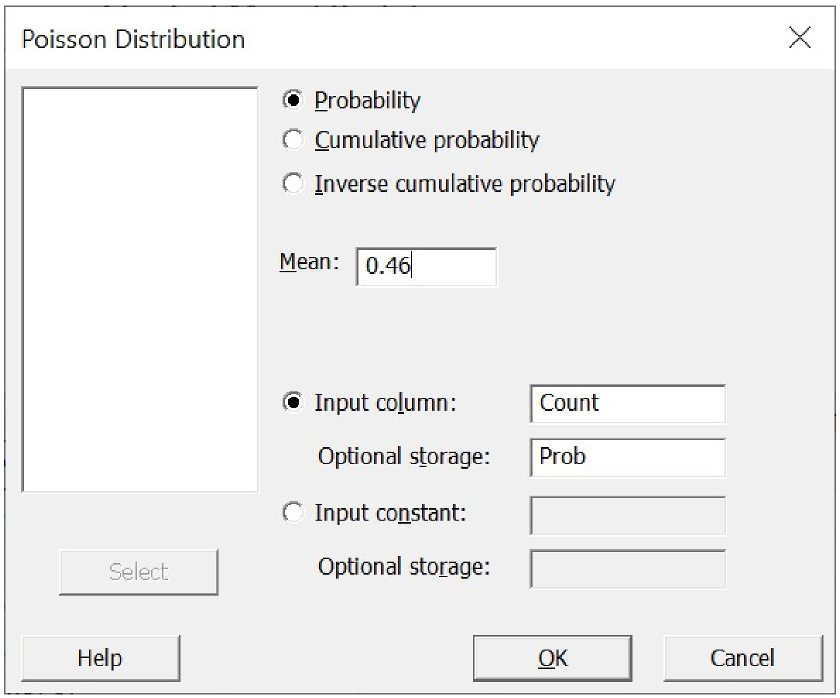
r	Pr(r)
0	9.550420e-01
1	4.393193e-02
2	1.010434e-03
3	1.549333e-05
4	1.781733e-07
5	1.639194e-09
6	1.256715e-11
7	8.258416e-14
8	4.748589e-16
9	2.427057e-18

Stats

mean: 0.046
variance: 0.046

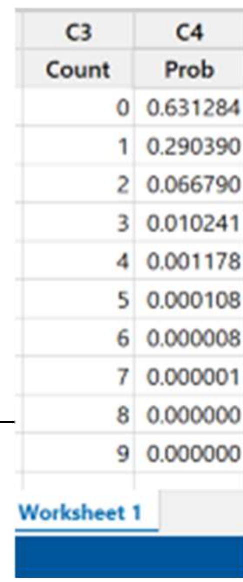
Poisson Calculation Example

Calc > Probability Distributions > Poisson

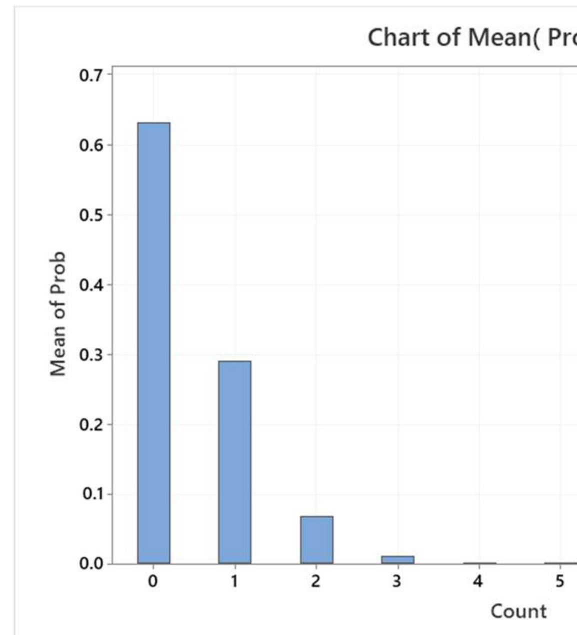


Poisson with mean = 0.046

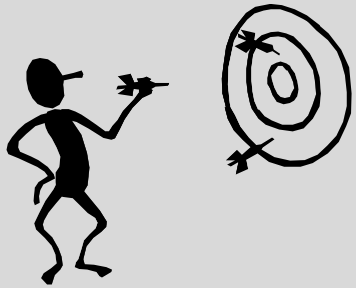
x	P(X = x)
0	0.955042
1	0.043932
2	0.001010
3	0.000015
4	0.000000
5	0.000000
6	0.000000
7	0.000000
8	0.000000
9	0.000000



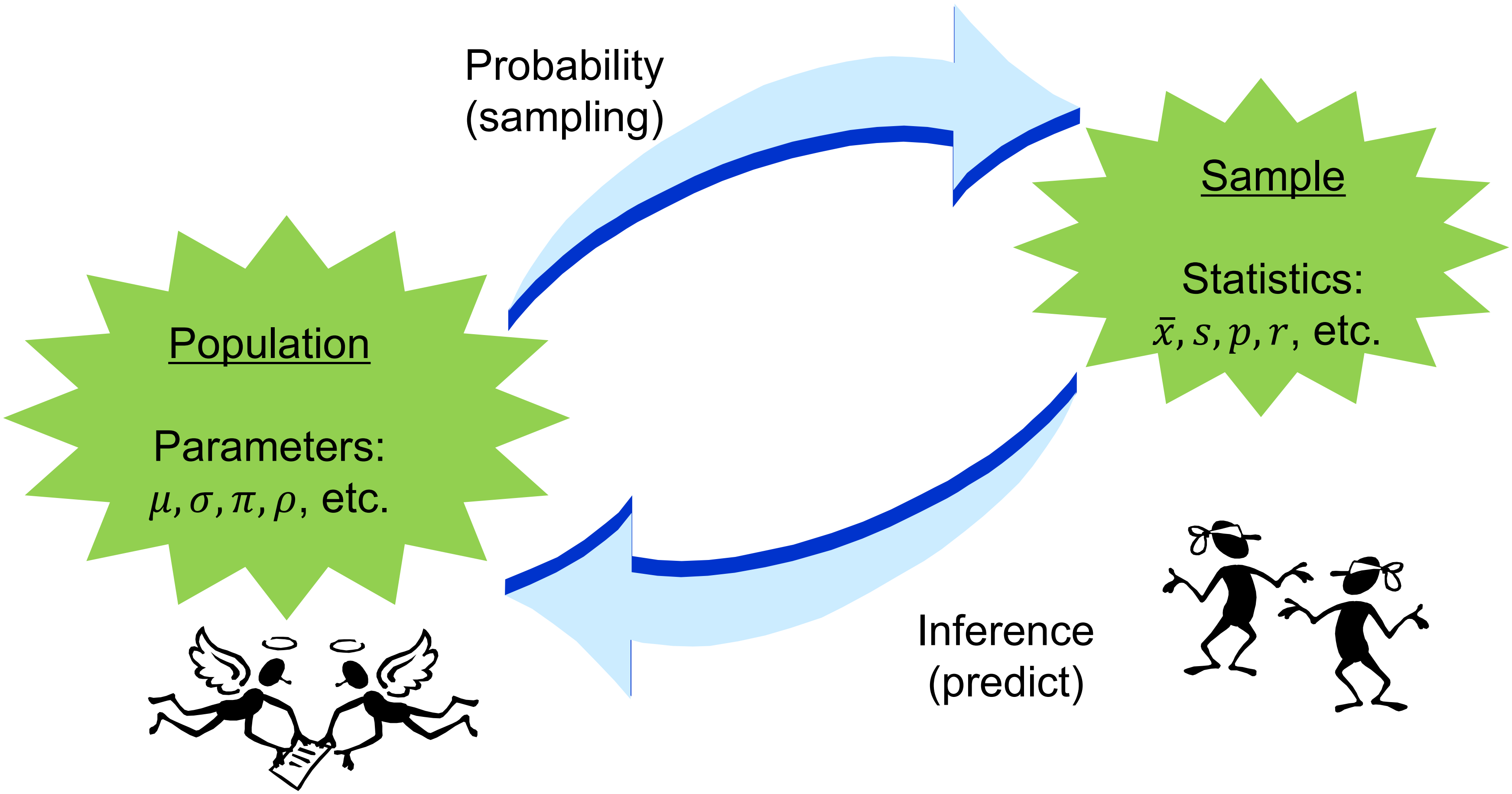
C3	C4
Count	Prob
0	0.631284
1	0.290390
2	0.066790
3	0.010241
4	0.001178
5	0.000108
6	0.000008
7	0.000001
8	0.000000
9	0.000000



Central Limit Theorem



Population Parameters & Sample Statistics



Central Limit Theorem – Why?

The ***Central Limit Theorem*** and ***Confidence Intervals*** are the fundamental tools used in making *inferential* statistical decisions.

The central limit theorem is the underlying concept of inferential statistics. It allows us to make inferences about a population characteristics based upon sample data.

Confidence Intervals are derived from the Central Limit Theorem.

The Central Limit Theorem

When independent random variables are added, their sum tends toward a normal distribution even if the original variables themselves are not normally distributed.

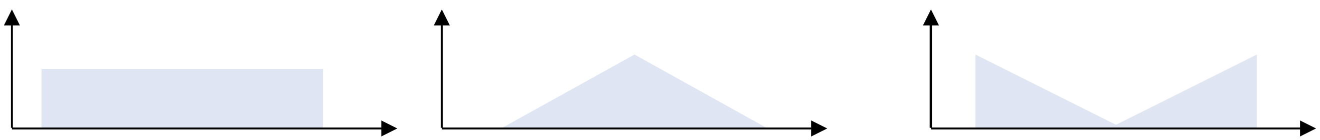
If $X_1, X_2, \ldots, X_n$ are n random samples drawn from a population with mean μ and variance σ^2, then the limiting distribution is a standard normal distribution,

$$Z = \lim_{n \to \infty} \sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right)$$

CLT – What Does It Do?

1. As the number of sample lots increases, the distribution of sample means becomes more normally distributed – *regardless of the underlying distribution of the individuals*.

Population



CLT – What Does It Do?

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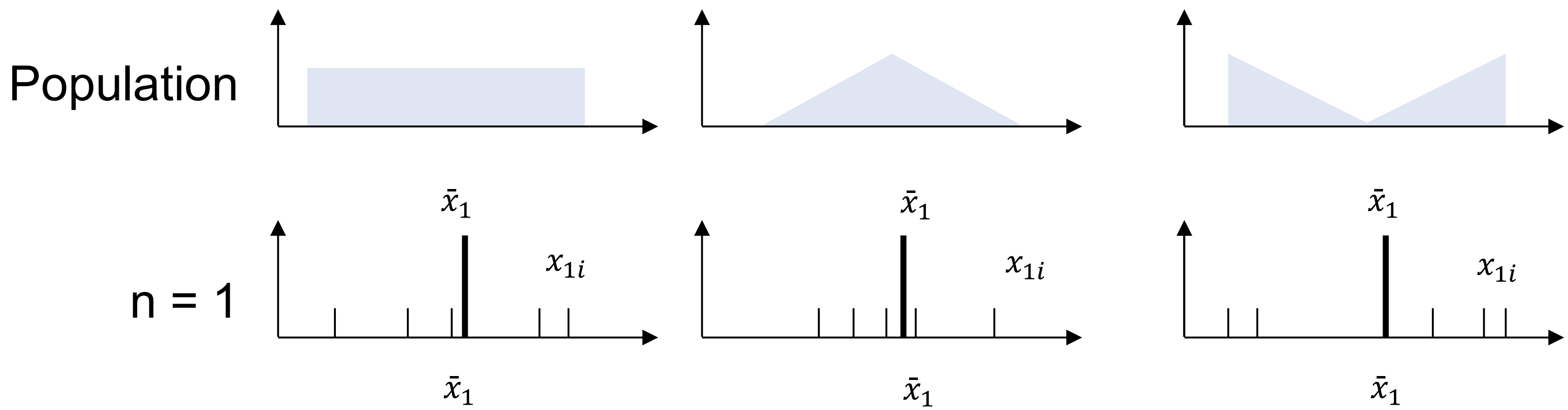

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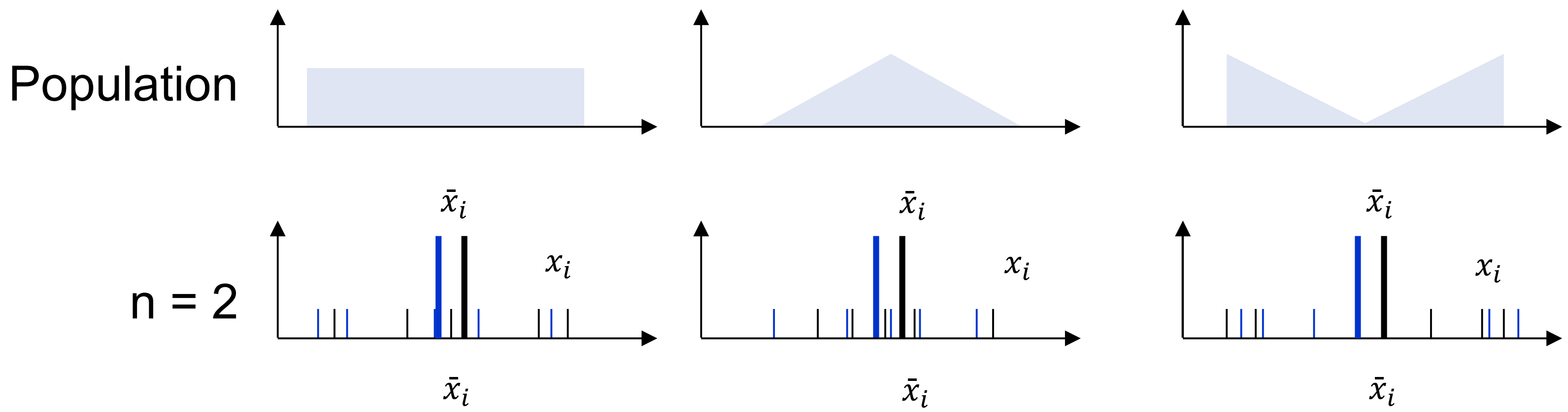

CLT – What Does It Do?

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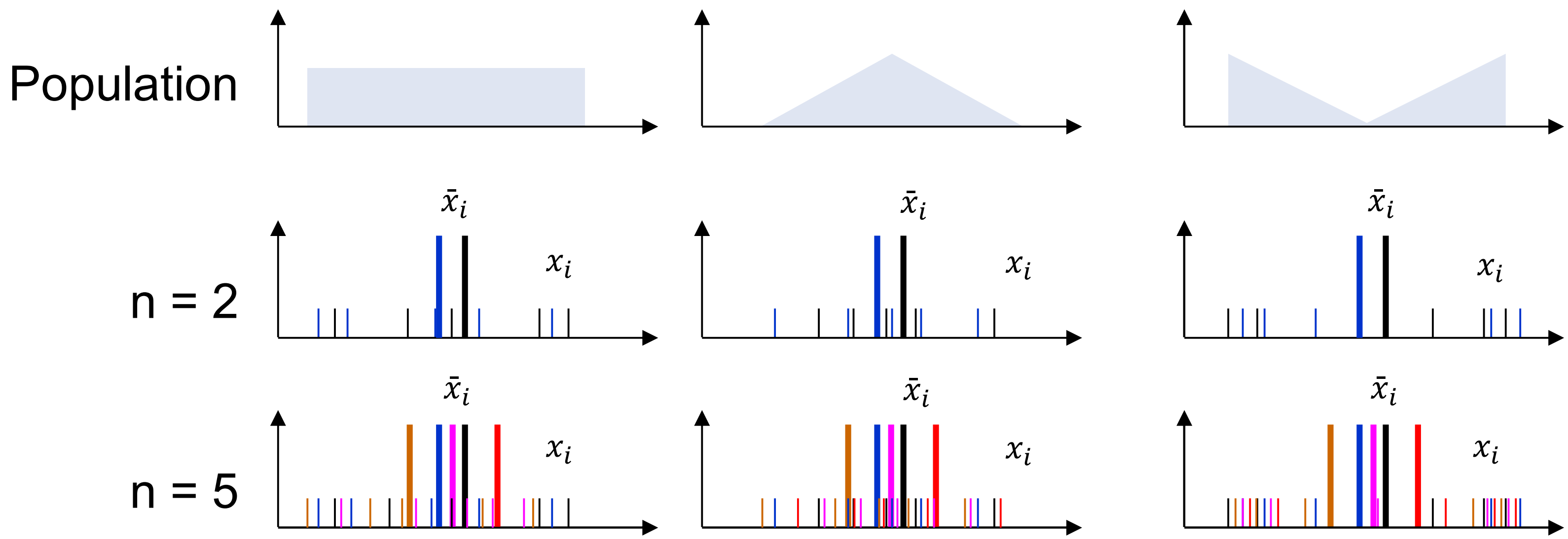
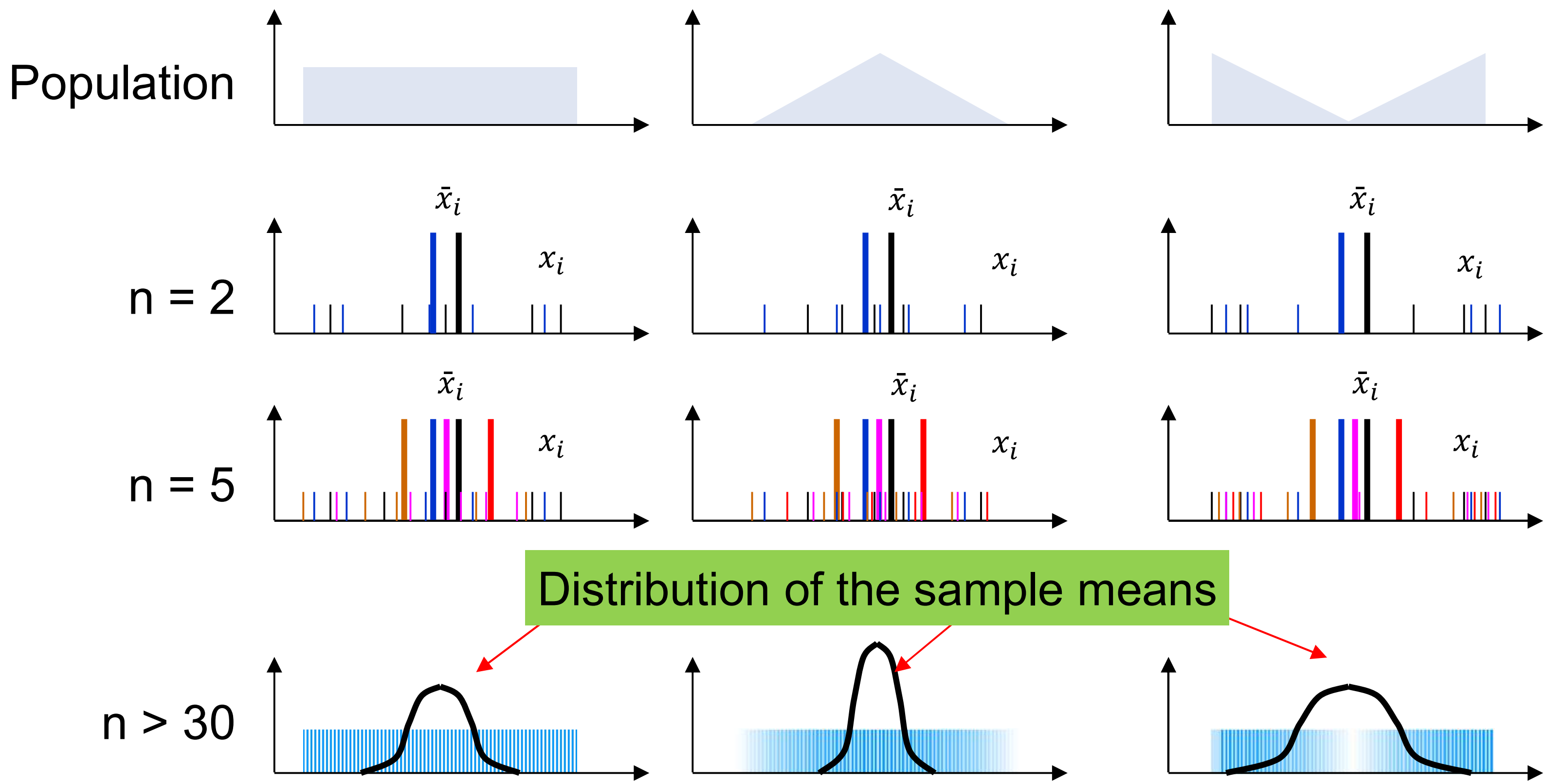


CLT – What Does It Do?

1. As the number of sample lots increases, the distribution of sample means becomes more normally distributed – *regardless of the underlying distribution of the individuals*.

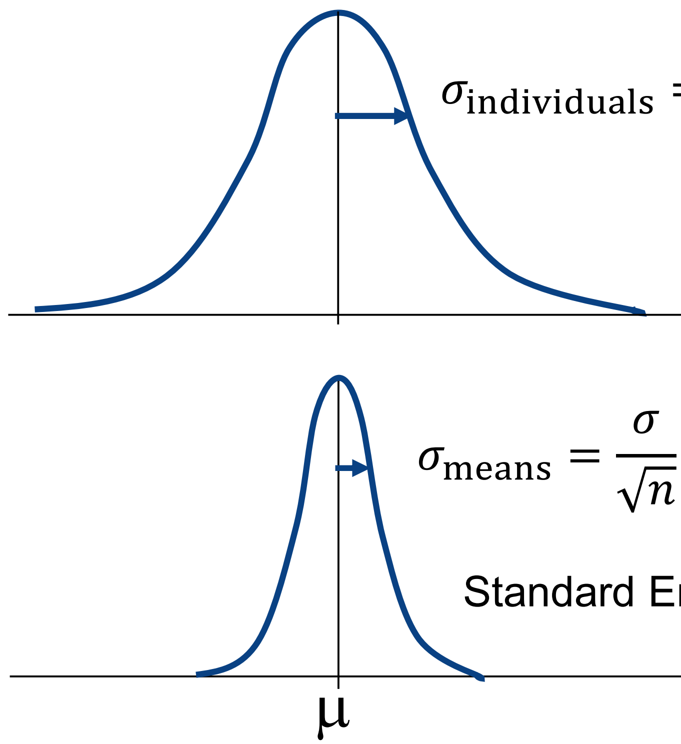


CLT – What Does It Do?

2. If multiple random samples of sample size n are taken from a normal distribution (mean μ and std dev σ), then the sample means will form a distribution with the same mean but with a smaller standard deviation.

$$\bar{x}_{\text{individuals}} = \bar{x}_{\text{means}} \approx \mu$$

$$\sigma_{\text{individuals}} = \sigma$$



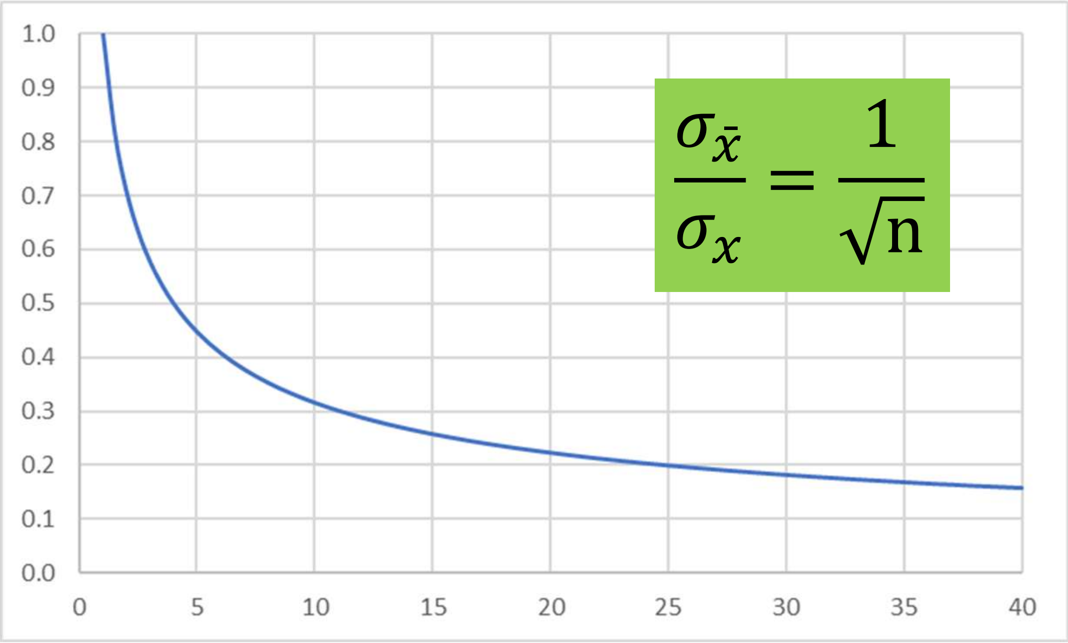
$$s_{\bar{x}} \approx \frac{s}{\sqrt{n}}$$

$$\sigma_{\text{means}} = \frac{\sigma}{\sqrt{n}}$$

Standard Error of the Mean

Standard Error and Sample Size

For approximately what sample size does the standard error of the mean begin to stabilize?

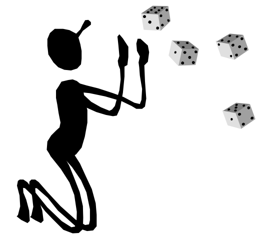


$$\frac{\sigma_{\bar{x}}}{\sigma_x}$$

$$\frac{\sigma_{\bar{x}}}{\sigma_x} = \frac{1}{\sqrt{n}}$$

Number of Samples n

CLT Exercise



Simulate the following scenario:

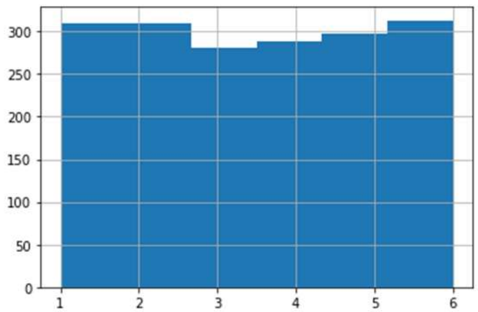
1. Throw 9 dice simultaneously and repeatedly 200 times.
2. Record the observations.
3. Create histograms and compute mean and standard deviation of
    a. The individual data
    b. The average of 4 dice.
    c. The average of 9 dice.

How do these distributions compare?

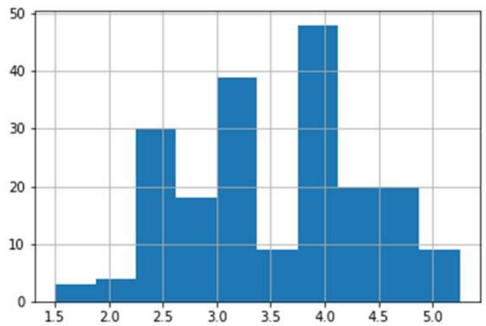
CLT Exercise - Python

```
import numpy as np
import pandas as pd

# generate 9 samples of 200 random die throws, as integers from 1-6:
data = np.random.randint(1, 7, size=(200,9))
df = pd.DataFrame(data, columns=list('123456789'))

hist = df.stack().hist(bins=6) # plot histogram. Should be uniform.

mean9 = df.mean(axis=1)
mean4 = df[['1','2','3','4']].mean(axis=1)

print('Average of individuals: ', df.stack().mean(),' and sigma of individuals:',df.stack().std())
print('Average of (average of 4 dies): ', mean4.mean(),' and sigma of (average of 4 dies):',mean4.std())
print('Average of (average of 9 dies): ', mean9.mean(),' and sigma of (average of 9 dies):',mean9.std())

print('Compare these sigmas with that exected by dividing by sqrt(n):')
print('Sigma of individuals   :',df.stack().std())
print('Sigma of individuals/2 :',df.stack().std()/2.0)
print('Sigma of individuals/3 :',df.stack().std()/3.0)

mean4.hist()
mean9.hist()
```

CLT Exercise - Python

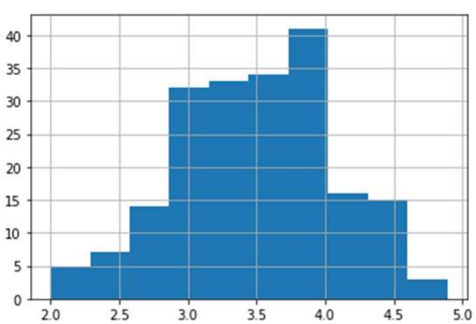



```
Average of individuals:  3.4955555555555557
and sigma of individuals: 1.732847240174108
```




```
Average of (average of 4 dies):  3.48125
and sigma of (average of 4 dies): 0.8342305680706731

Sigma of individuals/2 : 0.866423620087054
```

Sigma of average is smaller by ~$\sqrt{4}$




```
Average of (average of 9 dies):  3.495555555555556
and sigma of (average of 9 dies): 0.5812420304049765

Sigma of individuals/3 : 0.5776157467247026
```

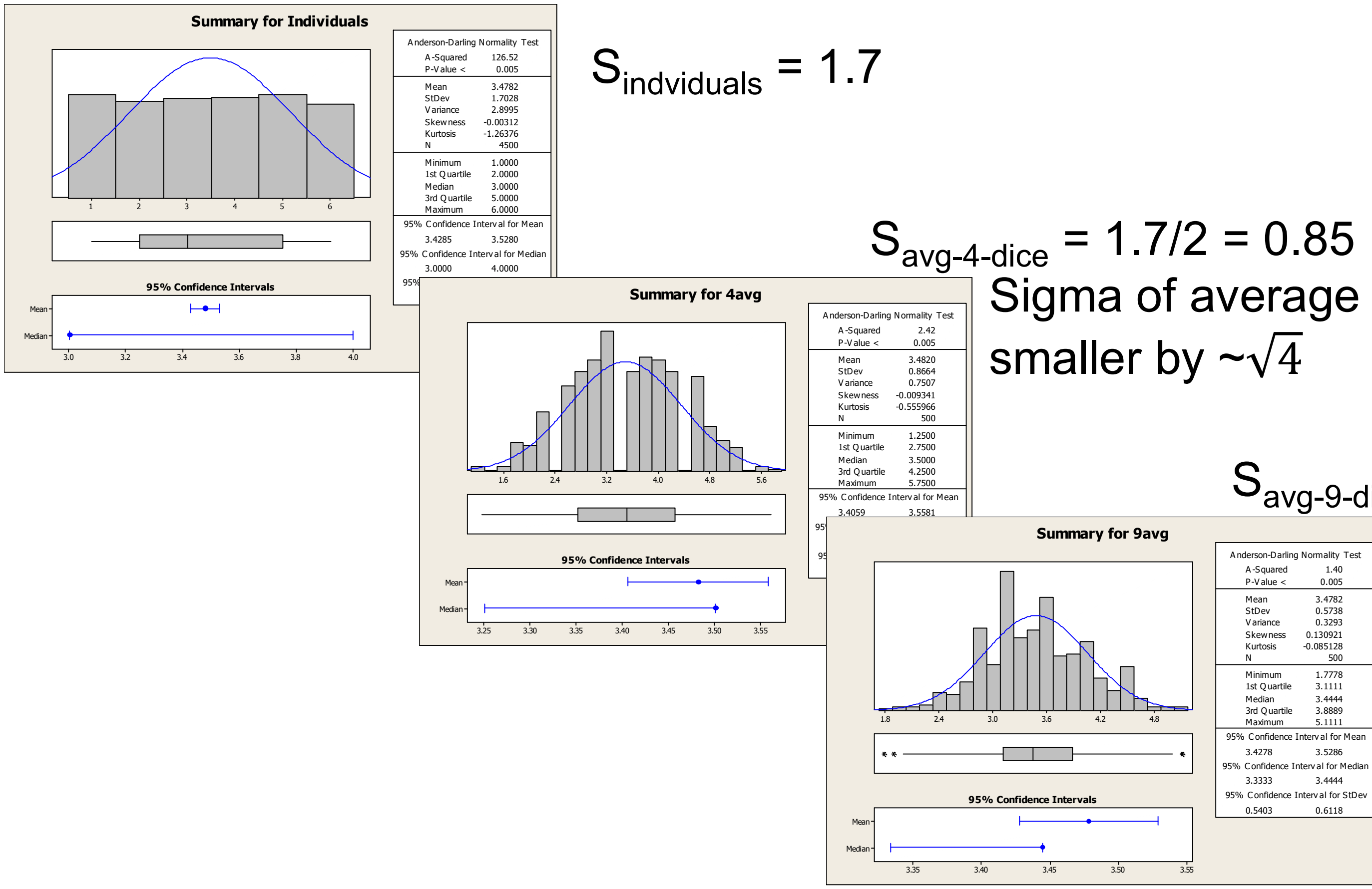
Sigma of average is smaller by ~$\sqrt{9}$

CLT Exercise - Minitab

This can be simulated using Minitab…

- Use Calc > Random Data > Integer
  to generate 500 rows of data in columns C1-C9
- Use Calc > Row Statistics (select mean)
  to create C10 & C11 as the mean of 9 & 4 rolls, respectively
- Use Data > Stack
  to stack C1-C9 data in Column C12, label as Individuals
- Use Stat > Graphical Summary
  to create histograms of C10, C11, and C12

CLT Exercise – Minitab



$S_{indviduals} = 1.7$

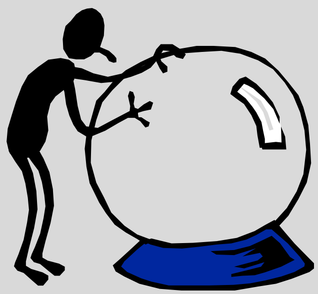
$S_{avg\text{-}4\text{-}dice} = 1.7/2 = 0.85$

Sigma of average is smaller by ~$\sqrt{4}$

$S_{avg\text{-}9\text{-}dice} = 1.7/3 = 0.56$

Sigma of average is smaller by ~$\sqrt{9}$

Confidence Intervals



Confidence Interval Example

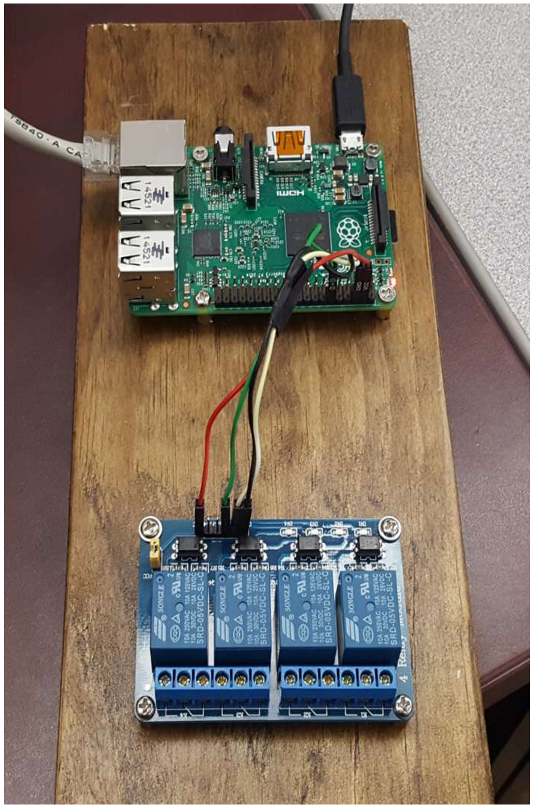
Suppose we have built a machine with a Raspberry-Pi controller that senses an input temperature and actuates a control based on the temperature.

The response time, in milliseconds, is a controller KPOV.

A new upgrade of the OS is installed.
Did it have any impact on controller performance?

- The mean response time previously was 4.7ms. Did it change?
- Within what range should we expect the average response time to fall 95% of the time (95% confidence interval for the mean response)?
- Within what range should we expect the standard deviation to fall 95% of the time (95% confidence interval for the standard deviation)?

These are questions we will answer with statistical inferencing.



Definitions: Point and Interval Estimates

A **Point Estimate** is a sample statistic such as a sample mean $\bar{x}$, a sample standard deviation s, or sample proportion defective p.

These statistics estimate the true population parameters μ, σ, π.

However, the point estimate from your sample will in fact be “wrong”. Why?

We can use the variability in the sample to create an interval that has some chance of containing the true population parameter.

An **Interval Estimate** is a range of values, with upper and lower limits, within which a population parameter has a certain probability of falling.

Concept: Point Estimate

Collect 3 samples of size n = 7 from a process characterized by a KPOV.

Sample 1	Sample 2	Sample 3
8000	8000	8300
7300	7500	7500
8000	7900	7600
8500	8300	8700
9500	9300	9500
9200	8900	9300
9500	9500	9000
$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$
8,571	8,485	8,557

Each of these 3 sample means is called a "point estimate"

Which value is correct?

How much confidence would you have in any of these values? In their grand average?

Concept: Point vs. Interval Estimate

Did we capture the true mean of the process this time?

Sample 1

8000

7300

8000

8500

9500

9200

9500

What is the KPOV mean?

Point Estimate Approach:

$\bar{x}$ = 8,571

Interval Estimate Approach:

The sample mean is $\bar{x}$ = 8,571 and the true mean is between 7780 and 9363 with 95% Confidence

Concept: Point vs. Interval Estimate

Collect samples of size n = 7 from each of the 2 machines with unknown mean performance. Do the machines differ?

Machine 1

8000
7000
8000
8500
9500
9000
9500

Are the means different?

Machine 2

8500
7000
9000
8500
9500
9000
9500

Point Estimate Approach:

Yes, they differ by
$|\bar{x}_1 - \bar{x}_2| = 214$

Interval Estimate Approach:

The best estimate is that they differ by $|\bar{x}_1 - \bar{x}_2| = 214$ and with 95% confidence, the true mean difference is within (-1257, 829). Since it might be zero, I can't confidently say they differ.

Confidence Level Choice

How confident can we afford to be and not-be in our estimates?

This value is our ***Confidence Level***.

We often use the ***risk level*** α to quantify confidence, as $CL = 1 - \alpha$.

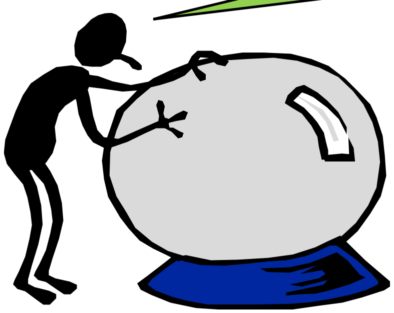
In most industries, we usually calculate to 90% or 95% confidence for our data, depending upon risk, 5-10% risk of being incorrect.

In the medical industry, we might calculate to 99.9% confidence for some of our calculations, 0.1% risk of being incorrect.

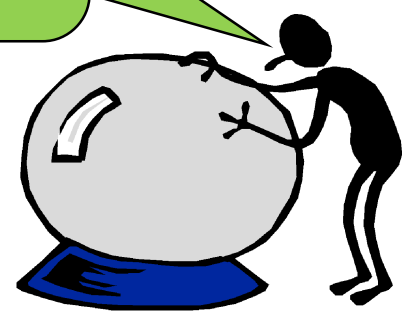
If we wanted to be 100% confident, how wide do you think our confidence intervals will be?

Confidence Level Choice

I’m 100% confident that
the yield is
between 0 and 100%!



I’m 100% confident that
the average is
between - ∞ and + ∞!



Interpretation of the CI:

A ***95% Confidence Interval*** can be interpreted as:

We are 95% certain that the true population parameter is inside the interval.

or

If we take 100 samples and compute a 95% confidence interval for each, 95 will have a CI that contains the true population parameter and 5 that do not.

or

If we take a new random sample, we are 95% certain the sample mean will be within the confidence interval.

Width of the CI

The width of a Confidence Interval depends on three things:

Population Spread (standard deviation)

- The higher the spread, the wider the CI

Sample size

- The higher the sample size the narrower the CI

Confidence Level (CL)

- The higher the CL the wider the CI

Incorrect Z-Score Thinking…

Back to the problem, the Response Time…

Within what range should we expect the average response time to fall 95% of the time (95% confidence interval for the mean response)?

Suppose we do 5 trials and collect data points with mean 4.392 and sigma of 0.33.

A 95% confidence is 1.96 sigma for normally distributed data.

So the confidence interval on the mean ought be ±1.96 sigma

$$\bar{x} - 1.96\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96\frac{s}{\sqrt{n}}$$

So the 95% confidence interval on the mean would be

$$4.103 \leq \mu \leq 4.681$$

But…

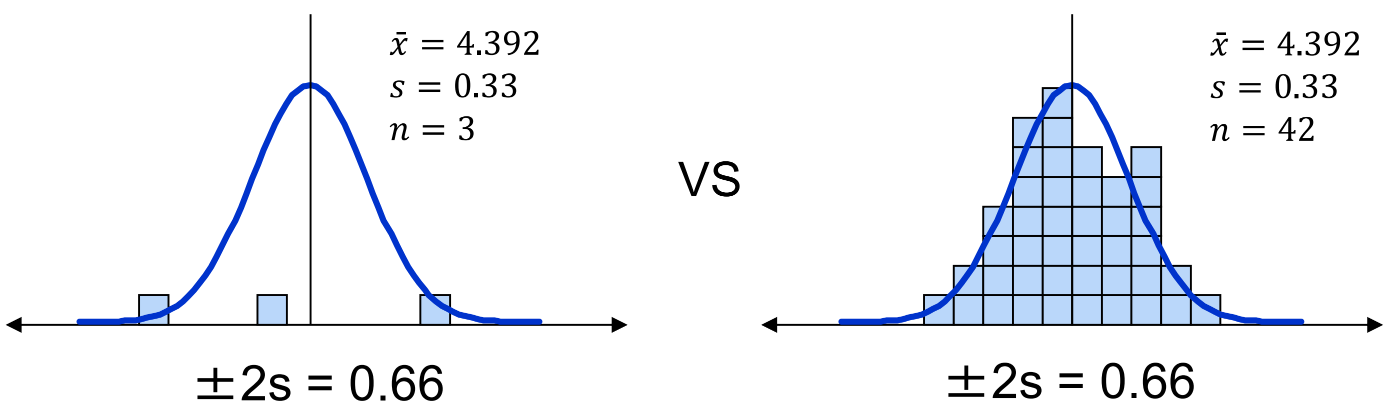
Sample Size has an Impact on Confidence

What if we had done only 3 trials?

What if we had done 42 trials?

Which would you have more confidence in your estimate of variance?

Using the normal distribution for computing these confidence intervals is not correct unless you are certain of the variance.



Confidence Interval for One Mean, σ Known

If $\bar{x}$ is the mean of an independent random sample of size n and given a known standard deviation σ, then the CI (two-tailed) for μ is

$$\text{CI of } \mu = \bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

This applies only if you know the population standard deviation.

- For example, if you have production data from previous years. Now you take a new sample today. Use the sample $\bar{x}$ but compute the CI using σ not s.

If σ is unknown and estimated by a sample s, then this confidence interval is not correct.

Confidence Interval for One Mean, σ Unknown

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This applies only if you know the population standard deviation.

- For example, if you have production data from previous years. Now you take a new sample today. Use the sample $\bar{x}$ but compute the CI using σ not s.

If σ is unknown and estimated by a sample s, then this confidence interval is not correct.

Instead, we need to use the ***t distribution***

$$\text{CI of } \mu = \bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}$$

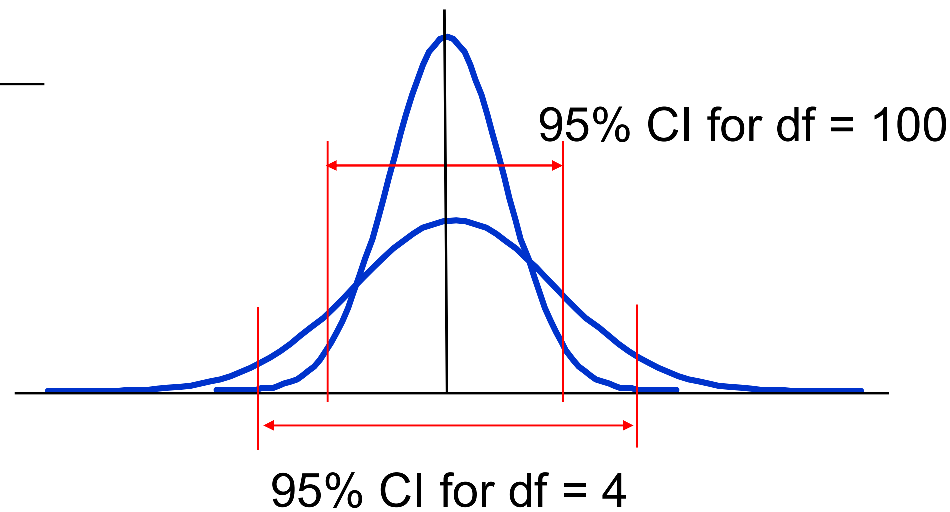
What is the t-Distribution?

The t-Distribution is a family of normal-shaped distributions but are further dependent on the sample size.

- The smaller the sample size, the wider the shape, compared to Z (standard normal)

Values of t for 95% CI's for different sample sizes:

Sample	t-value
5	2.78
10	2.26
20	2.09
30	2.05
100	1.98
1000	1.96



What is the t-Distribution?

The t-Distribution is a family of normal-shaped distributions but are further dependent on the sample size.

- Parameterized by the **degrees of freedom** $df = \nu = n - 1$, the number of values in the calculation of a statistic that are free to vary. The number of data points minus the number of parameters.

The probability density function $\varphi(t, \nu)$ is given by

$$\varphi(t,\nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

where $\Gamma()$ is the gamma function.

In Excel: For a two-sided calculation,

$\varphi(t,\nu) =$ =t.dist.2d(t, ν)

and $\varphi^{-1}(P,\nu) =$ =t.inv.2d(P, ν)

In Python:

$\varphi(t,\nu) =$ `= t.cdf(`t, ν`)`

and $\varphi^{-1}(P,\nu) =$ `= t.ppf(`P, ν`)`

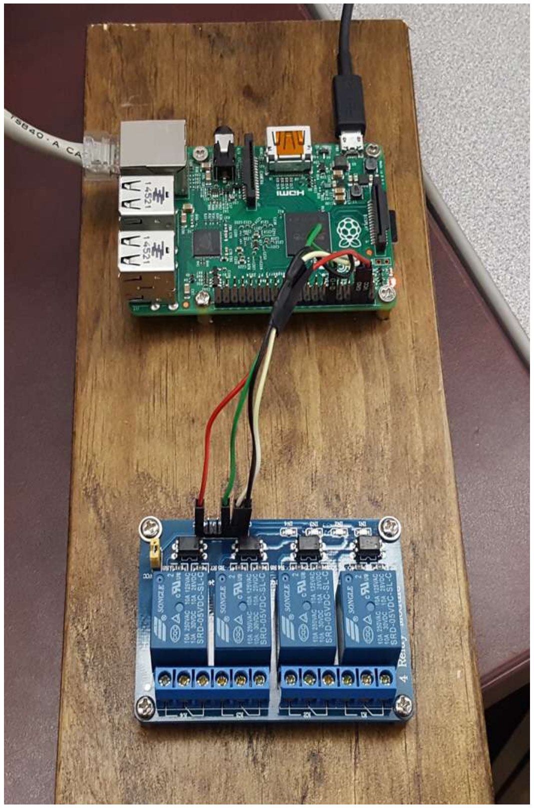
Confidence Interval Example

Back to the problem, the Response Time…

Within what range should we expect the average response time to fall 95% of the time (95% confidence interval for the mean response)?

Within what range should we expect the standard deviation to fall 95% of the time (95% confidence interval for the standard deviation)?

The mean response time was 4.7ms.
Did it change?



Confidence Interval Example

Suppose we want to determine the 95% CI for the mean from a sample of 25 response times. We run 25 trials and get sample mean = 4.392 msec, sample sigma = 0.3303 msec

We can calculate the CI by hand as follows: $\bar{x} - t_{\frac{\alpha}{2},n-1}\left(\frac{s}{\sqrt{n}}\right) \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2},n-1}\left(\frac{s}{\sqrt{n}}\right)$

First, $t_{\frac{\alpha}{2},n-1} = t_{5\%/2,\,24}$: in Excel this is =tinv.2d(0.95,25) = 2.064

Substituting, $4.392 - 2.064 \times \frac{0.3303}{\sqrt{25}} \leq \mu \leq 4.392 + 2.064 \times \frac{0.3303}{\sqrt{25}}$

$$4.392 - 0.1363 \leq \mu \leq 4.392 + 0.1363$$

$$4.256 \leq \mu \leq 4.528$$

We are 95% sure that our true response time is between 4.256 and 4.528 ms. We're also taking a 5% chance that is wrong. Is the response time changed from the earlier 4.7ms?

EXERCISE: Confidence Interval Example

Confidence interval calculations are provided in statistical code libraries typically using the data collected itself as the inputs.

Suppose the data was as shown.

```
import mqr

a = np.array([4.23, 3.98, 4.36, 3.86, 4.12, 4.23, 4.36,
              4.44, 3.98, 4.06, 3.79, 4.35,4.20,4.57,
              4.82,4.44, 4.94,4.68,4.57,4.44,4.36,4.82,
              4.74,5.01,4.45])

mqr.inference.mean.confint_1sample(a, conf=0.95)
```

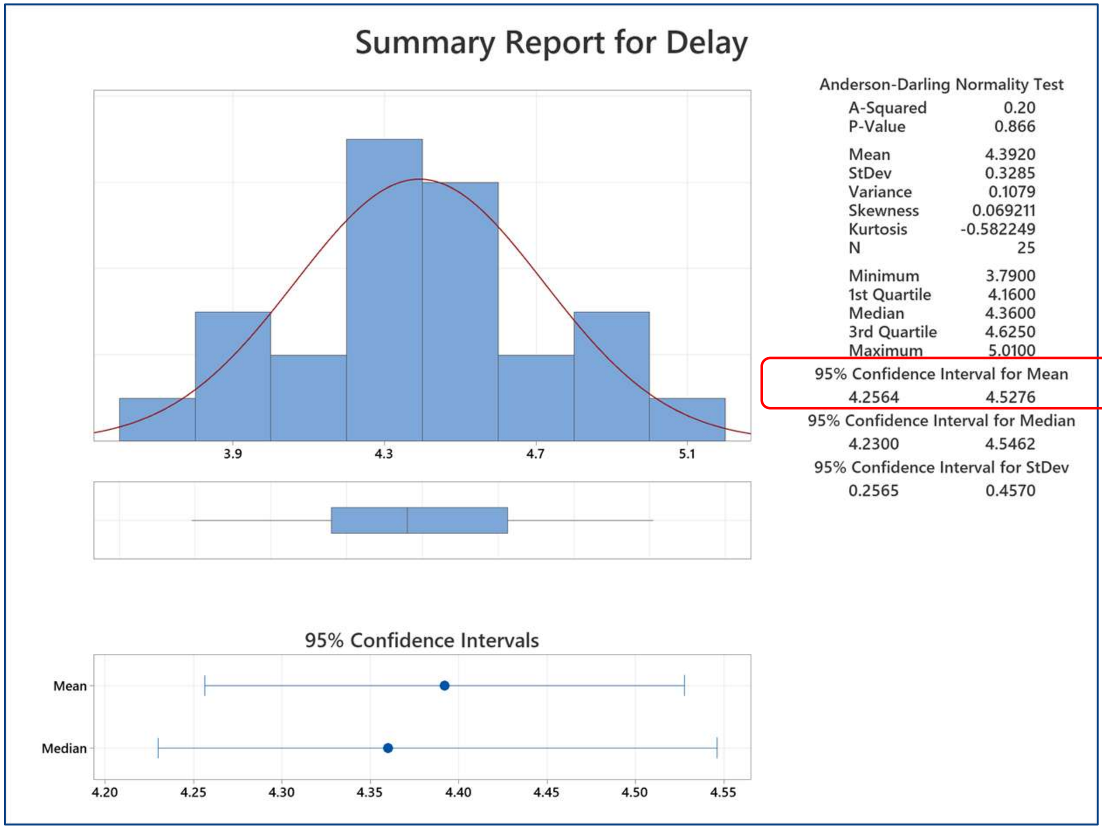
Confidence Interval
mean

value	[2.5%	97.5%]
4.392	4.25641	4.52759

4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45

Confidence Interval Example

Minitab:



4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45

Confidence Interval Calculations

There are three basic questions posed which we answer with statistical confidence intervals.

Is there a difference between a sample and a target?

i.e., is the target in the confidence interval?

Is there a difference in a sample before and after some process?

i.e., is zero change in the confidence interval?

Is there a difference between two samples?

i.e., is zero difference in the confidence interval?

Confidence Interval Calculations

Continuous Set of real valued measurements (KPI inspection values)	Pass/Fail Set of binary measurements. (Defective unit or not)	Count Integer measurement. Number of defects per unit.
Is the mean (variance) of a sample different from a target?	Is the scrap rate of a sample different from a target?	Is the defect rate of a sample different from a target?
Is the mean (variance) of a sample before before a process different from the mean after?	Is the scrap rate of a sample before a process different from the scrap rate after?	Is the defect rate of a sample before a process different from the scrap rate after?
Is the mean (variance) of one sample different from a second?	Is the scrap rate of one sample different from a second?	Is the defect rate of one sample different from a second?

Confidence Intervals for Parameters – Python

Parameter	Sampling Distribution	Confidence Interval	Python Function
Mean μ (σ known)	Z	$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$	`st.norm.interval`
Mean μ, (σ unknown)	t	$\bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}$	`mqr.inference.mean.confint_1sample`
Paired data difference in means $\mu - \mu'$	t	$\bar{d} \pm t_{\frac{\alpha}{2},n-1} \frac{s_d}{\sqrt{n}}$	`mqr.inference.mean.confint_paired`
Two means $\mu_1 - \mu_2$	t	$\bar{x}_1 - \bar{x}_1 \pm t_{\frac{\alpha}{2},n_1+n_2-1} s_p$	`mqr.inference.mean.confint_2sample`
Variance σ^2	χ^2	$s\sqrt{\frac{n-1}{\chi^2_{1-\frac{\alpha}{2},n-1}}} \leq \sigma \leq s\sqrt{\frac{n-1}{\chi^2_{\frac{\alpha}{2},n-1}}}$	`mqr.inference.stddev.confint_1sample`
Variance Ratio $\frac{\sigma_1^2}{\sigma_2^2}$	F	$\frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2},(n_2-1)(n_1-1)} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} F_{1-\frac{\alpha}{2},(n_2-1)(n_1-1)}$	`mqr.inference.variance.confint_2sample`

Confidence Intervals for Parameters – Python

Parameter	Sampling Distribution	Confidence Interval	Python Function
Rate ρ	Poisson	$r \pm \frac{1}{2nt}\chi^2_{2s,1-\frac{\alpha}{2}}$	`mqr.inference.rate.confint_1sample`
Two rates $\rho_1 - \rho_2$	Poisson	$r_1 - r_2 \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{s_1}{n_1 t_1} + \frac{s_2}{n_2 t_2}}$	`mqr.inference.rate.confint_2sample`
Proportion π	Binomial	$\frac{2xF_{\frac{\alpha}{2},\nu_1,\nu_2}}{2(n-x+1) + 2xF_{\frac{\alpha}{2},\nu_1,\nu_2}} \le \pi \le \frac{2(x+1)F_{\frac{\alpha}{2},\nu_1,\nu_2}}{2(n-x) + 2(x+1)F_{\frac{\alpha}{2},\nu_1,\nu_2}}$	`mqr.inference.proportion.confint_1sample`
Two proportions $\pi_1 - \pi_2$	Binomial	$(p_1 - p_2) \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	`mqr.inference.proportion.confint_2sample`

Confidence Intervals for Parameters – Minitab

Parameter	Sampling Distribution	Confidence Interval	Minitab
Mean μ (σ known)	Z	$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$	Stat > Basic Stats > 1 Sample Z...
Mean μ , (σ unknown)	t	$\bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}$	Stat > Basic Stats > 1 Sample t...
Paired data difference in means $\mu - \mu'$	t	$\bar{d} \pm t_{\frac{\alpha}{2},n-1} \frac{s_d}{\sqrt{n}}$	Stat > Basic Stats > Paired t...
Two means $\mu_1 - \mu_2$	t	$\bar{x}_1 - \bar{x}_1 \pm t_{\frac{\alpha}{2},n_1+n_2-1} s_p$	Stat > Basic Stats > 2 Sample t...
Variance σ^2	χ^2	$s\sqrt{\frac{n-1}{\chi^2_{1-\frac{\alpha}{2},n-1}}} \leq \sigma \leq s\sqrt{\frac{n-1}{\chi^2_{\frac{\alpha}{2},n-1}}}$	Stat > Basic Stats > 1 Variance...
Variance Ratio $\frac{\sigma_1^2}{\sigma_2^2}$	F	$\frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2},(n_2-1)(n_1-1)} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} F_{1-\frac{\alpha}{2},(n_2-1)(n_1-1)}$	Stat > Basic Stats > 2 Variances...

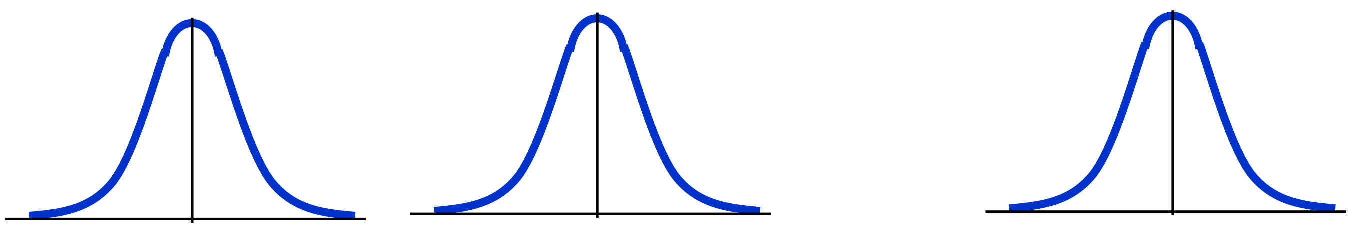
Confidence Intervals for Parameters – Minitab

Parameter	Sampling Distribution	Confidence Interval	Minitab
Rate ρ	Poisson	$r \pm \frac{1}{2nt}\chi^2_{2s,1-\frac{\alpha}{2}}$	Stat > Basic Stats > One Sample Poisson Rate…
Two rates $\rho_1 - \rho_2$	Poisson	$r_1 - r_2 \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{s_1}{n_1 t_1} + \frac{s_2}{n_2 t_2}}$	Stat > Basic Stats > Two Sample Poisson Rates…
Proportion π	Binomial	$\frac{2xF_{\frac{\alpha}{2},v_1,v_2}}{2(n-x+1)+2xF_{\frac{\alpha}{2},v_1,v_2}} \le \pi \le \frac{2(x+1)F_{\frac{\alpha}{2},v_1,v_2}}{2(n-x)+2(x+1)F_{\frac{\alpha}{2},v_1,v_2}}$	Stat > Basic Stats > One Proportion…
Two proportions $\pi_1 - \pi_2$	Binomial	$(p_1 - p_2) \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	Stat > Basic Stats > Two Proportions…

Confidence Interval for the Difference in Means

The difference of two means which are normally distributed is itself normally distributed. So:

- We can continue to use either the standard normal distribution (Z) when the σ's are known,
- or the t distribution, when the σ's are not known, to create the confidence intervals



$$\mu_1 \quad - \quad \mu_2 \quad = \quad \mu_1 - \mu_2$$

Confidence Interval on $\mu_1 - \mu_2$ for Paired Observations

If d and s_d are the sample mean and sample standard deviation of the difference of n random pairs of normally distributed measurements,

then CI (two-tailed) on the difference in means $\mu_d = \mu_1 - \mu_2$ is:

$$\text{CI of } \mu_d = \bar{d} \pm t_{\frac{\alpha}{2}, n-1} \frac{s_d}{\sqrt{n}}$$

Example: CI for Paired Means

Suppose plastic parts were inspected for tensile strength. Then suppose they were sterilized through a UV process and then the parts re-inspected for tensile strength. The data is shown. Did the UV process affect tensile strength?

Part	S_{before} (MPa)	S_{after} (MPa)
1	34.18	33.59
2	33.25	33.13
3	32.68	32.84
4	31.03	32.01
5	40.49	36.75
6	39.81	36.41
7	38.22	35.61
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Let $\mu_d = \mu_{\text{After}} - \mu_{\text{Before}}$

$$\text{CI of } \mu_d = \bar{d} \pm t_{\frac{\alpha}{2},n-1} \frac{s_d}{\sqrt{n}}$$

Computing using Minitab or Python,

$$\text{CI of } \mu_d = (-2.39, -0.04)$$

Zero is not in the confidence interval. Therefore, to 95% confidence, we can say the mean shifted lower.

C. I. for the Difference in Means, σ's Unknown and Unequal

If σ_1 and σ_2 are unknown and unequal

Two-tailed CI for the difference in means $(\mu_1 - \mu_2)$ is

$$\text{CI of } \mu_1 - \mu_2 = (\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2},\nu}\sqrt{\frac{s_1^2}{n_1} - \frac{s_2^2}{n_2}}$$

where ν is the pooled degree of freedom

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1+1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2+1}} - 2$$

Example: Confidence Interval for $\mu_1 - \mu_2$

Suppose a sample of 10 plastic parts were inspected for tensile strength, and from two difference batches of raw material. Is there a difference in tensile strength of the two batches?

Part	S1 (MPa)	S2 (MPa)
1	34.18	33.59
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7	38.22	35.61
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8	33.23	33.12
9	37.52	35.26
10	33.90	33.45

$$\text{CI of } \mu_1 - \mu_2 = (\bar{x}_1 - \bar{x}_2) - t_{\frac{\alpha}{2},\nu}\sqrt{\frac{s_1^2}{n_1} - \frac{s_2^2}{n_2}}$$

Computing using Minitab or Python,

$$\text{CI of } \mu_1 - \mu_2 = (-1.29, 3.72)$$

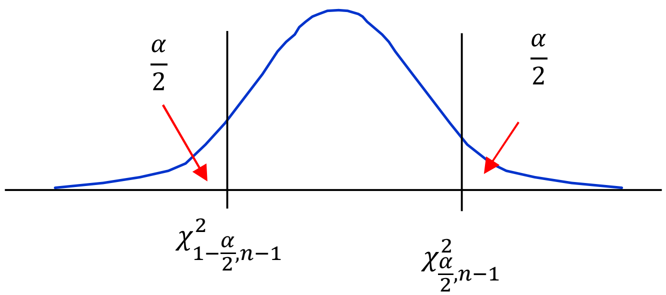
Zero is in the confidence interval. Therefore, to 95% confidence, we can say the mean has not changed.

Confidence Interval for σ

Consider a random sample of a normally distributed population.

Recall the sample variance is

$$s^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$



We can derive $\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 = \frac{(n-1)s^2}{\sigma^2}$, but this defines a χ^2 Chi-Square distribution. The random variable $\frac{(n-1)s^2}{\sigma^2}$ follows a χ^2 distribution with (n-1) degree of freedom. The two-tailed CI for σ is then

$$\chi^2_{\frac{\alpha}{2},n-1} \le \frac{(n-1)s^2}{\sigma^2} \le \chi^2_{1-\frac{\alpha}{2},n-1}$$

Re-arranging

$$s\sqrt{\frac{n-1}{\chi^2_{1-\frac{\alpha}{2},n-1}}} \le \sigma \le s\sqrt{\frac{n-1}{\chi^2_{\frac{\alpha}{2},n-1}}}$$

Example: Confidence Interval for σ

Consider the response time example, and the sample of 25 data points that was taken.

The sample standard deviation s = 0.3285.

The degrees of freedom (n-1) = 25-1 = 24.

The 95% (α = 0.05) Confidence Interval for Sigma is calculated as:

$$s\sqrt{\frac{n-1}{\chi^2_{1-\frac{\alpha}{2},n-1}}} \leq \sigma \leq s\sqrt{\frac{n-1}{\chi^2_{\frac{\alpha}{2},n-1}}}$$

4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45

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$$s\sqrt{\frac{n-1}{\chi^2_{1-\frac{\alpha}{2},n-1}}} \leq \sigma \leq s\sqrt{\frac{n-1}{\chi^2_{\frac{\alpha}{2},n-1}}}$$

$$0.3285\sqrt{\frac{25-1}{\chi^2_{1-\frac{0.05}{2},24}}} \leq \sigma \leq 0.3285\sqrt{\frac{25-1}{\chi^2_{\frac{0.05}{2},24}}}$$

$$0.3285\sqrt{\frac{24}{\chi^2_{0.975,24}}} \leq \sigma \leq 0.3285\sqrt{\frac{24}{\chi^2_{0.025,24}}}$$

$$0.3285\sqrt{\frac{24}{39.36}} \leq \sigma \leq 0.3285\sqrt{\frac{24}{12.40}}$$

$$0.2546 \leq \sigma \leq 0.4570$$

Confidence Interval for the Ratio of Variances

Consider two independent normal populations with unknown variance σ_1^2 and σ_2^2.

If s_1^2 and s_2^2 are the sample variances of random samples of size n_1 and n_2 respectively from the two populations, then the CI of their ratio is given by

$$\frac{s_1^2}{s_2^2} F_{\frac{\alpha}{2},(n_2-1)(n_1-1)} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{s_1^2}{s_2^2} F_{1-\frac{\alpha}{2},(n_2-1)(n_1-1)}$$

where $F_{\frac{\alpha}{2},(n_2-1)(n_1-1)}$ and $F_{\frac{\alpha}{2},(n_2-1)(n_1-1)}$ are the lower and upper $\frac{\alpha}{2}$ percentage points of the F distribution.

Example: CI for the Ratio of Variances

Consider two samples of data as shown. Do they exhibit the same variance?

sample1	sample2
4.23	5.84
3.98	5.62
4.36	5.96
3.86	5.51
4.12	5.75
4.23	5.84
4.36	5.96
4.44	6.04
3.98	5.62
4.06	5.69
3.79	5.44
4.35	5.95
4.20	5.82
4.57	6.16
4.82	6.38
4.44	6.04
4.94	6.49
4.68	6.25
4.57	6.16
4.44	6.04
4.36	5.96
4.82	6.38
4.74	6.31
5.01	6.56
4.45	6.05

Example: CI for the Ratio of Variances

Consider two samples of data as shown. Do they exhibit the same variance?

Computing using Minitab or Python, we can compute

$$s_1 = 0.328, s_2 = 0.299$$

and

$$s_1/s_2 = 0.911$$

and the confidence interval on this ratio as

$$(0.728, 1.65)$$

The ratio value of 1.0 is in the confidence interval. Therefore, to 95% confidence, we can say the variance has not changed.

sample1	sample2
4.23	5.84
3.98	5.62
4.36	5.96
3.86	5.51
4.12	5.75
4.23	5.84
4.36	5.96
4.44	6.04
3.98	5.62
4.06	5.69
3.79	5.44
4.35	5.95
4.20	5.82
4.57	6.16
4.82	6.38
4.44	6.04
4.94	6.49
4.68	6.25
4.57	6.16
4.44	6.04
4.36	5.96
4.82	6.38
4.74	6.31
5.01	6.56
4.45	6.05

Example: CI for Paired Means – Python

Suppose plastic parts were inspected for tensile strength. Then suppose they were sterilized through a UV process and then the parts re-inspected for tensile strength. The data is shown. Did the UV process affect tensile strength?

Part	S_{before} (MPa)	S_{after} (MPa)
1	34.18	33.59
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8	33.23	33.12
9	37.52	35.26
10	33.90	33.45

Example: CI for Paired Means – Minitab

Suppose plastic parts were inspected for tensile strength. Then suppose they were sterilized through a UV process and then the parts re-inspected for tensile strength. The data is shown. Did the UV process affect tensile strength?

Part	S_{before} (MPa)	S_{after} (MPa)
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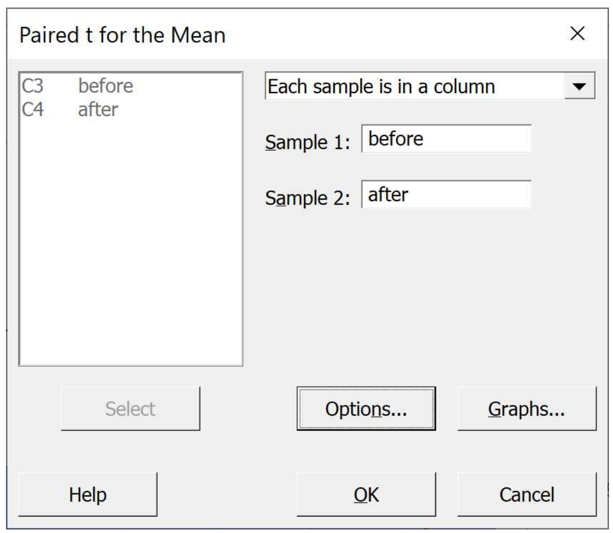
6	39.81	36.41
7	38.22	35.61
8	33.23	33.12
9	37.52	35.26
10	33.90	33.45

Stat > Basic Stats > Paired t…



Estimation for Paired Difference

Mean μ_	StDev	SE Mean	95% CI for difference
1.214	1.643	0.520	(0.039, 2.389)

μ_difference: population mean of (before - after)

Example: Confidence Interval for $\mu_1 - \mu_2$

Suppose a sample of 10 plastic parts were inspected for tensile strength, and from two difference batches of raw material. Is there a difference in tensile strength of the two batches?

Part	S1 (MPa)	S2 (MPa)
1	34.18	33.59
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4	31.03	32.01
5	40.49	36.75
6	39.81	36.41
7	38.22	35.61
8	33.23	33.12
9	37.52	35.26
10	33.90	33.45

```
import numpy as np, statsmodels.stats.api as sm

X1 = np.array([34.18,33.25,32.68,31.03,
              40.49,39.81,38.22,33.23,37.52,33.90])
X2 = np.array([33.59,33.13,32.84,32.01,36.75,
              36.41,35.61,33.12,35.26,33.45])

mqr.inference.mean.confint_2sample(X1, X2, conf=0.95,
                                   pooled=False, bounded='both')
```

Confidence Interval
difference between means (independent)
95% (t)

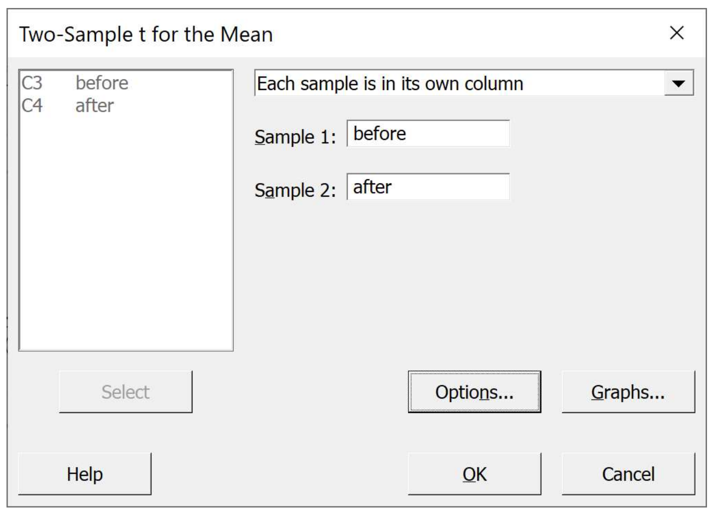
value	lower	upper
1.214	-1.29399	3.72199

Example: Confidence Interval for $\mu_1 - \mu_2$

Suppose a sample of 10 plastic parts were inspected for tensile strength, and from two difference batches of raw material. Is there a difference in tensile strength of the two batches?

Part	S1 (MPa)	S2 (MPa)
1	34.18	33.59
2	33.25	33.13
3	32.68	32.84
4	31.03	32.01
5	40.49	36.75
6	39.81	36.41
7	38.22	35.61
8	33.23	33.12
9	37.52	35.26
10	33.90	33.45

Stat > Basic Stats > 2 Sample t…



Estimation for Difference

Difference	95% CI for Difference
1.21	(-1.30, 3.73)

Example: Confidence Interval for σ

Using Python: Suppose the data was as shown.

```
import numpy as np, scipy.stats as st

arr = np.array([4.23, 3.98, 4.36, 3.86, 4.12, 4.23, 4.36,
              4.44, 3.98, 4.06, 3.79, 4.35,4.20,4.57,
              4.82,4.44, 4.94,4.68,4.57,4.44,4.36,4.82,
              4.74,5.01,4.45])
ci = mqr.inference.stddev.confint_1sample(arr, conf=0.95, bounded='above’)
display(ci)
```

Confidence Interval
standard deviation
95% (chi2)

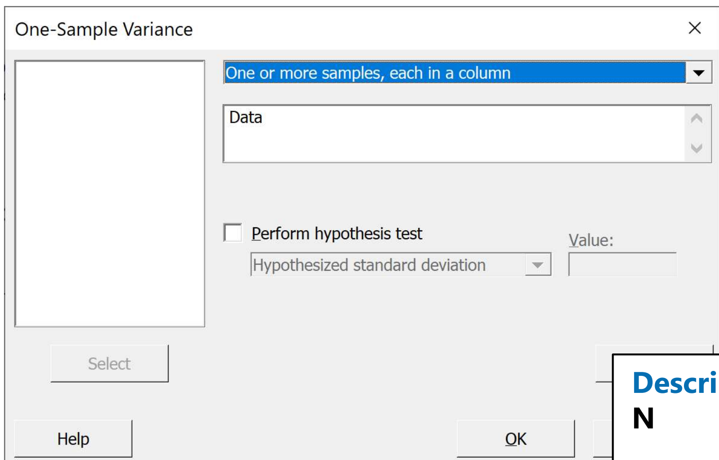
value	lower	upper
0.328481	0	0.43243

4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45

Example: Confidence Interval for σ

Using Minitab.

Stat > Basic Stats > 1 Variance…



Descriptive Statistics

N	StDev	Variance	95% CI for σ using Bonett	95% CI for σ using Chi-Square
25	0.328	0.108	(0.264, 0.443)	(0.256, 0.457)

4.23
3.98
4.36
3.86
4.12
4.23
4.36
4.44
3.98
4.06
3.79
4.35
4.20
4.57
4.82
4.44
4.94
4.68
4.57
4.44
4.36
4.82
4.74
5.01
4.45

Example: CI for the Ratio of Variances

Using Python: Suppose the data was as shown.

```
import numpy as np, scipy.stats as st

alpha = 0.05                 # significance level = 5%
n1 = len(sample1)            # sample sizes
n2 = len(sample2)            # sample sizes
s21 = np.var(sample1, ddof=1)    # sample variance
s22 = np.var(sample2, ddof=1)    # sample variance
df1 = n1 - 1                  # degrees of freedom
df2 = n2 - 1                  # degrees of freedom
upper = s21 / s22 * st.f.ppf(1 - alpha / 2, df2, df1)
lower = s21 / s22 * st.f.ppf(alpha / 2, df2, df1)
lower = np.sqrt(lower)
upper = np.sqrt(upper)
print('StDev1 =',np.std(sample1,ddof=1), ', StDev2 =',
np.std(sample2,ddof=1))
print('s2/s1 Ratio = ', np.sqrt(s22/s21))
print("(", lower, "," , upper , ")")
```

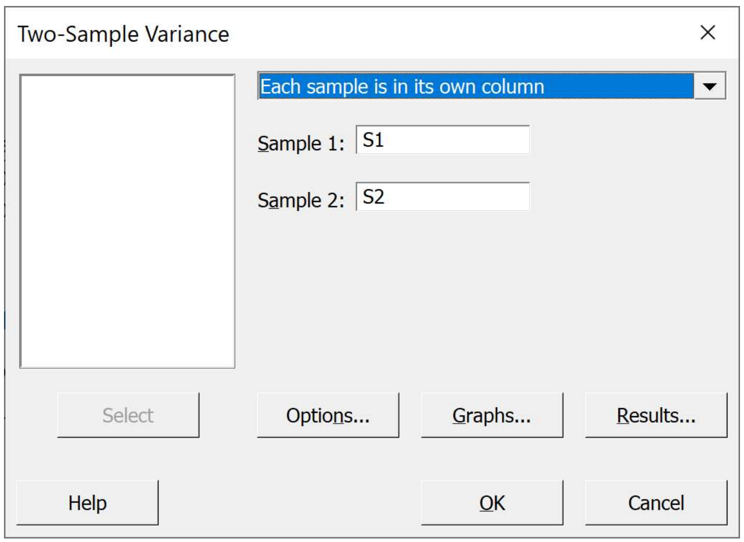
```
StDev1 = 0.32848135411313684 , StDev2 = 0.29932590933629505
s2/s1 Ratio =  0.9112417054673979
( 0.7284883812218992 , 1.653142130518071 )
```

sample1	sample2
4.23	5.84
3.98	5.62
4.36	5.96
3.86	5.51
4.12	5.75
4.23	5.84
4.36	5.96
4.44	6.04
3.98	5.62
4.06	5.69
3.79	5.44
4.35	5.95
4.20	5.82
4.57	6.16
4.82	6.38
4.44	6.04
4.94	6.49
4.68	6.25
4.57	6.16
4.44	6.04
4.36	5.96
4.82	6.38
4.74	6.31
5.01	6.56
4.45	6.05

Example: CI for the Ratio of Variances

Using Minitab

Stat > Basic Stats > 2 Variances…



Ratio of Standard Deviations

Estimated Ratio	**95% CI for Ratio using F**
1.09740	(0.728, 1.653)

sample1	sample2
4.23	5.84
3.98	5.62
4.36	5.96
3.86	5.51
4.12	5.75
4.23	5.84
4.36	5.96
4.44	6.04
3.98	5.62
4.06	5.69
3.79	5.44
4.35	5.95
4.20	5.82
4.57	6.16
4.82	6.38
4.44	6.04
4.94	6.49
4.68	6.25
4.57	6.16
4.44	6.04
4.36	5.96
4.82	6.38
4.74	6.31
5.01	6.56
4.45	6.05

Confidence Interval for a Proportion

Confidence intervals can also be constructed for fraction defective, π (or any proportion such as yields), where

- x = number of defective occurrences
- n = sample size
- p = x/n proportion defective in sample

The Binomial Distribution is the correct distribution to describe proportions. The confidence interval is

$$\frac{2xF_{\frac{\alpha}{2},\nu_1,\upsilon_2}}{2(n-x+1)+2xF_{\frac{\alpha}{2},\nu_1,\upsilon_2}} \le \pi \le \frac{2(x+1)F_{\frac{\alpha}{2},\nu_1,\upsilon_2}}{2(n-x)+2(x+1)F_{\frac{\alpha}{2},\nu_1,\upsilon_2}}$$

Example: CI for a Proportion

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 3 defectives were observed. Is the yield better than 95%?

Computing $\frac{2xF_{\frac{\alpha}{2},\nu_1,\nu_2}}{2(n-x+1)+2xF_{\frac{\alpha}{2},\nu_1,\nu_2}} \leq \pi \leq \frac{2(x+1)F_{\frac{\alpha}{2},\nu_1,\nu_2}}{2(n-x)+2(x+1)F_{\frac{\alpha}{2},\nu_1,\nu_2}}$ using Minitab or Python, the result is

$$(0.0, 0.063)$$

To 95% confidence, the defective rate could be as high as 6.3%. That is, the yield could be as low as 93.7%. Therefore, no, we cannot confidently say the yield is better than 95%.

C. I. for the Difference in 2 Proportions

Assume that the normal approximation to the Binomial distribution applies

Then for the difference in the true proportions $(\pi_1 - \pi_2)$

$$\text{CI of } (\pi_1 - \pi_2) = (p_1 - p_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Example: CI for the Difference in 2 Proportions

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 10 defectives were observed. Yesterday, 3 were observed in a sample of 100. Has the yield changed?

Computing $(p_1 - p_2) \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$ with $p_1 = \frac{3}{100}$ and $p_2 = \frac{10}{100}$ using Minitab or Python, the result is

$$(-0.147, 0.0005)$$

The confidence interval contains zero, no change. Therefore, despite the 3X higher rate observed in the sample, no, we cannot confidently say the yield has changed. The sample size is too low.

Confidence Interval for a Rate

Confidence intervals can also be constructed for the rate of occurrence, ρ (such as a defect rate), where

- s = number of defect occurrences
- t = length of a sample
- n = sample size
- $r = s/(nt)$ = defect rate in the sample

The Poisson Distribution is the correct distribution to describe rates, and

$$\text{CI of } \rho = r \pm \frac{1}{2nt} \chi^2_{2s, 1-\frac{\alpha}{2}}$$

`

Example: CI for a Rate

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. What is the defect rate confidence interval?

Computing $r \pm \frac{1}{2nt}\chi^2_{2s,1-\frac{\alpha}{2}}$ with s=90∗0+7∗1+3∗2=13, n=100 and t=1, using Minitab or Python, the result is

$$(6.92, 22.2)$$

The average defect rate is somewhere between 7 and 22 defects per 100 units. That is, between 0.07 and 0.22 defects per unit (less than one defect on each unit). The best estimate is the sample value, 13 defects per 100 units, or 0.13 defects per unit.

Confidence Interval for the difference of two rates

Confidence intervals can also be constructed for the difference of two rate of occurrences, ($\rho_1 - \rho_2$) (such as defect rates), where

- s = number of defect occurrences
- t = length of a sample
- n = sample size
- $r = s/(nt)$ = defect rate in the sample

The Poisson Distribution is the correct distribution to describe rates, and

$$\text{CI of } \rho_1 - \rho_2 = r_1 - r_2 \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{r_1}{n_1 t_1} + \frac{r_2}{n_2 t_2}}$$

Example: CI for the difference of two rates

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. Yesterday there were 85 with no defects, 10 with 1 defect and 5 with 2 defects. Was there a change?

Here s_1=90$*$0+7$*$1+3$*$2=13 and s_2=85$*$0+10$*$1+5$*$2=20, n=100 and t=1, using Minitab or Python, the result per unit is

$$(-0.18, 0.04)$$

The average defect rate per unit change is somewhere between going down by 0.18 defects or going up by 0.04 defects per unit. This confidence interval include zero, or zero changes in defects per unit, so we cannot confidently say the rate has changed.

Summary

The mathematics of probability govern uncertainty in manufacturing and quality control.

Data can be of several types: defectives count, defects count, continuous measurements.

We often use distribution functions to describe the probability of occurrence of events, defined using population parameters.

Statistics are estimates of parameters using samples.

The average is a statistic describing the central tendency, the variance is a statistic describing the spread.

The central limit theorem proves that statistics converge to parameters with more samples.

Confidence intervals provide intervals to estimate population parameters.

Example: CI for a Proportion – Python

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 3 defectives were observed. Is the yield better than 95%?

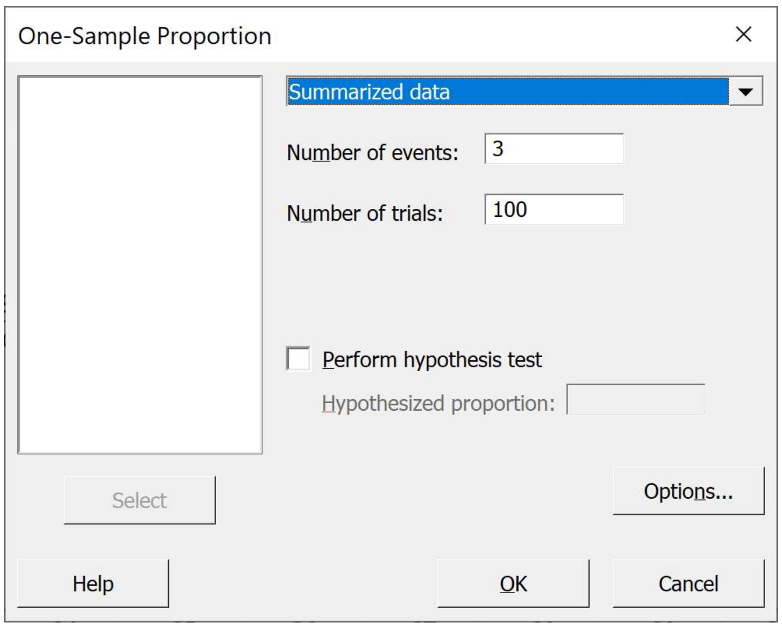
```
import numpy as np, statsmodels.stats.api as sm

count = 3
nobs = 100
ci_low, ci_upp = sm.proportion_confint(count, nobs, alpha=0.05)
ci_low, ci_upp
```

```
(0.0, 0.06343448095637183)
```

Example: CI for a Proportion – Minitab

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 3 defectives were observed. Is the yield better than 95%?

Stat > Basic Stats > 1 Proportion…



Descriptive Statistics

N	Event	Sample p	95% CI for p
100	3	0.030000	(0.006230, 0.085176)

Example: CI for the Difference in 2 Proportions

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 10 defectives were observed. Yesterday, 3 were observed in a sample of 100. Has the yield changed?

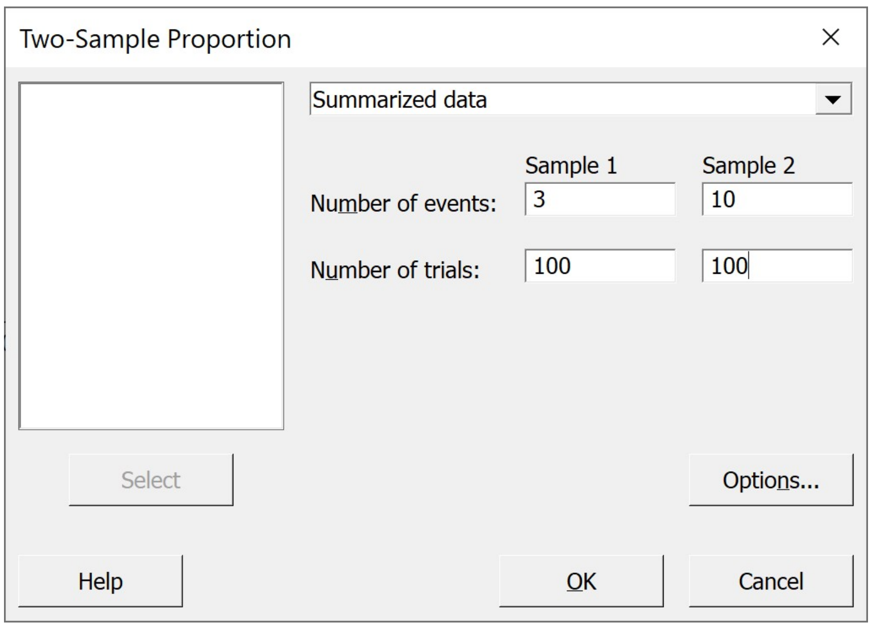
```
import numpy as np
from statsmodels.stats.proportion import confint_proportions_2indep

count1 = 3
nobs1 = 100
count2 = 10
nobs2 = 100
ci_low, ci_upp = confint_proportions_2indep(count1, nobs1,
                                            count2, nobs2,alpha=0.05)
ci_low, ci_upp
```

```
(-0.14694241634222716, 0.000546376597066106l)
```

Example: CI for the Difference in 2 Proportions

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 10 defectives were observed. Yesterday, 3 were observed in a sample of 100. Has the yield changed?

Stat > Basic Stats > 2 Proportions…



Estimation for Difference

Difference	95% CI for Difference
-0.07	(-0.137640, -0.002360)

CI based on normal approximation

Example: CI for a Rate

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. What is the defect rate CI?

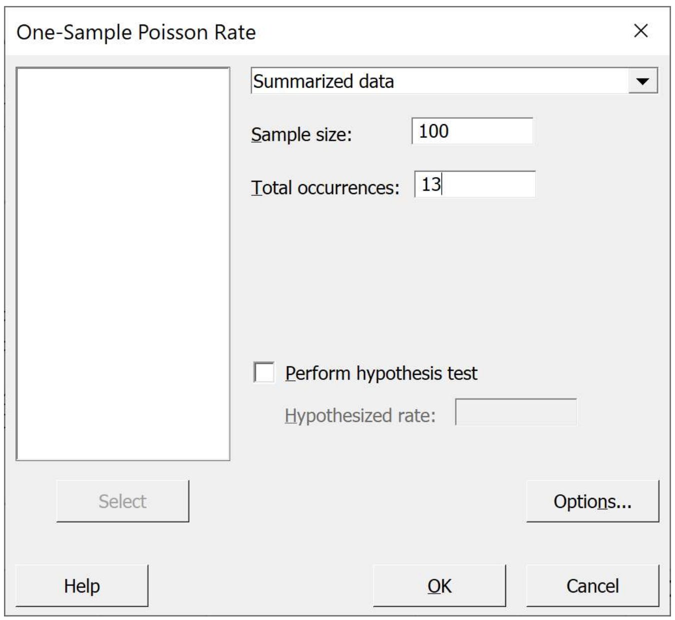
```
import numpy as np
from scipy.stats.distributions import chi2

count = 13
nobs = 100
alpha = 0.05

lb = 0.5 * chi2.ppf(alpha/2,df=2*count)
ub = 0.5 * chi2.ppf(1-alpha/2,df=2*count+2)
lb, ub
```

```
(6.921952491003801, 22.230395918158873)
```

Example: CI for a Rate

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. What is the defect rate?

Stat > Basic Stats > 1 Sample Poisson Rate…



Descriptive Statistics

N	Total Occurrences	Sample Rate	95% CI for λ
100	13	0.13	(0.0692195, 0.222304)

Example: CI for the difference of two rates

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. Yesterday there were 85 with no defects, 10 with 1 defect and 5 with 2 defects. Was there a change?

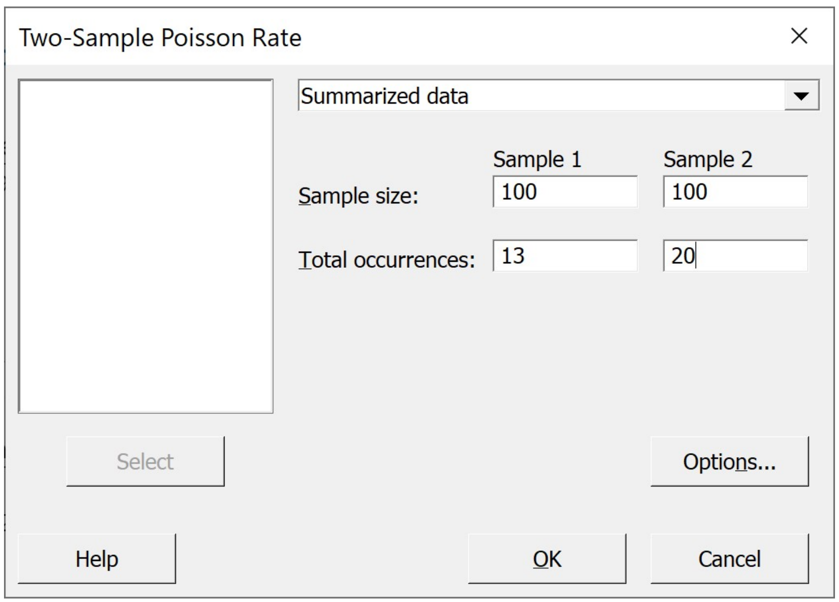
```
import numpy as np
from scipy.stats.distributions import norm

count1 = 13
t1 = 1
nobs1 = 100
count2 = 20
t2 = 1
nobs2 = 100
alpha = 0.05

r1 = count1/t1/nobs1
r2 = count2/t2/nobs2
print('Sample 1 rate =', r1, ', Sample 2 rate =', r2)
print('Difference in sample rates =', r1-r2)

s = np.sqrt(r1/t1/nobs1 + r2/t2/nobs2)
lb = r1 - r2 + norm.ppf(alpha/2)*s
ub = r1 - r2 - norm.ppf(alpha/2)*s
lb, ub
```

```
Sample 1 rate = 0.13 , Sample 2 rate = 0.2
Difference in sample rates = -0.07

(-0.18259135894148, 0.04259135894148)
```

Example: CI for the difference of two rates

Suppose a sample of 100 parts were randomly sampled and inspected from today's production, and 90 had no defects, 7 had 1 defect and 3 had 2 defects each. Yesterday there were 85 with no defects, 10 with 1 defect and 5 with 2 defects. Was there a change?

Stat > Basic Stats > 2 Sample Poisson Rate…



Estimation for Difference

Estimated Difference	95% CI for Difference
-0.07	(-0.182591, 0.0425914)